Performance Funding and Equity of Access to Public Universities

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Abstract

State appropriations to public universities have historically been determined by enrollment, but an alternative is "performance funding" (PF), which is based on outcomes metrics such as degree completions. Advocates argue that PF reduces an incentive to over-enroll, while critics argue PF incentivizes universities to admit fewer under-represented minority (URM) applicants, as they are less likely to graduate. I develop a theoretical framework in which a social planner sets a funding rule and universities make enrollment decisions based on competing institutional priorities, including the funding rule. The theory predicts that changes in level of funding, not structure of the rule, affects enrollment. To assess model predictions, I use synthetic controls to estimate enrollment changes at public four-year universities in Ohio, a state that has used PF since 2009. Consistent with the theory, I find that enrollment level has a positive correlation with level of funding, not the timing of the switch to PF. The switch to PF coincides with a long-run decrease in funding levels as well as a 1 percentage point decrease in long-run enrollment. This decrease was disproportionately borne by Black students: the proportion of Black students among first-time cohorts decreased by 1.13 percentage points.

Keywords: efficiency and equity, education and inequality, education policy

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1 Introduction

U.S. state governments have historically appropriated funds to public universities based on enrollment headcount (enrollment funding, "EF"), but an alternative is to reward universities based on performance metrics such as degree completions (performance funding, "PF"). PF initially gained traction among policymakers during the 1980s, when the proportion of high school graduates enrolling in post-secondary educational institutions was increasing, yet the college completion rate was decreasing (Bound, Lovenheim, and Turner 2010). Because this trend could indicate that universities admit students who are unlikely to graduate—which is undesirable due to the high opportunity cost of attending university, especially for students who do not complete a degree—advocates for PF argue that it reduces an unintended incentive for universities to deliberately over-enroll students who are unlikely to graduate. However, critics argue that PF also incentivizes universities to admit fewer applicants whose observable characteristics correlate with a lower probability of degree completion. If so, then switching from EF to PF may disproportionately restrict college accessibility to students who already belong to an under-represented minority (URM), as URM students are less likely to graduate than non-URM students.[1]

33 states used some form of PF in 2020, but the majority of states do not consistently use PF and/or only use PF to determine small amounts of funding (Ortagus, Rosinger, and Kelchen 2021). An exception is Ohio, as the state has used PF to determine the majority of its state appropriations since 2009.[2] While Ohio is an important case study, Ohio's experiences with PF may not generalize well to other states. For example, Ohio has a proportionately smaller Hispanic population than the national average, so state policymakers with larger Hispanic populations may not be able to directly extrapolate from Ohio's experiences. Theoretical modeling is therefore useful to understanding how funding rules affect public university enrollment more generally, and can provide a framework that policymakers in many different contexts can adapt and use to inform their decision-making.

Researchers have traditionally framed PF as part of a principal-agent problem in which state policymakers (the principal) use funding rules to induce public universities (the agents) to take costly actions that align with the principal's desired outcome, increased degree completions. The principal-agent framework captures a potential moral hazard problem that PF introduces: while it is intended to make universities invest in programs/policies to increase student degree completions, it also introduces a perverse incentive to lower completion requirements or otherwise "game" the funding rules. For example, Goldring, Jacob, Kreisman,

[1] Causey, Lee, Ryu, Scheetz, and Shapiro (2022) report that the six-year completion rate at four-year public universities for the Fall 2016 cohort is 68% overall, but only 50.2% for Black students and 57.1% for Latino/a students.

[2] Another notable exception is Tennessee, which has used PF to determine the majority of its state funding since 2010. However, the switch to PF coincides with the expansion of higher education programs that also affect enrollment at public four-year institutions, making it challenging to untangle the enrollment effects of concurrent policy changes. I discuss the case of Tennessee separately at the end of the paper, in Appendix E.

and Ricks (2024) find that when Michigan changed funding of its Career and Technical Education programs to reward student progression, students didn't change course-taking behavior, but administrators strategically rearranged curricula around the new funding rule.

However, this paper focuses exclusively on how different funding rules affect a public university's *enrollment decision*, which is fundamentally different from the traditional principal-agent problem. Neither the cost nor the benefit of making an enrollment decision depend on the funding rule. Therefore, the funding rule cannot be a mechanism that state policymakers use to incentivize universities to take on a costly action. Rather, funding rules may change institutional *preferences* over applicants, because some applicants are expected to bring more funding than others. So while the post-enrollment effects of PF on universities are best modeled as a principal-agent problem, the enrollment effects are better modeled as a matching problem that incorporates funding rules into an institution's preferences over which students to enroll. This paper is intended to help fill the conceptual modeling gap.

In this paper, I theoretically and empirically characterize the effects of switching to PF on first-time enrollment at public universities. Characterizing how policy changes affect enrollment is important because all higher education outcomes are downstream of being admitted—therefore, understanding how funding affects outcomes in higher education begins by first understanding how it affects which students are admitted. In Section 2, I review the existing literature. Next, I provide an overview on public universities and state funding in Ohio in Section 3.

In Section 4, I then develop a flexible theoretical framework in which a state policymaker (or social planner, SP) sets a funding rule and universities make enrollment decisions based on competing institutional priorities, including the funding rule chosen. State policymakers and universities consider different costs and benefits to enrolling students, and these differences cause universities to enroll different sets of students than the policymaker would, if the policymaker could directly intervene in the admissions process. Since direct intervention is infeasible, a state policymaker can instead use an appropriation funding rule as a tool to affect a university's enrollment incentives. The theoretical model predicts that there are many enrollment-based and performance-based funding rules that can realign the university's enrollment incentives with the state policymaker's enrollment incentives (Theorem 1). That is, PF is not *intrinsically* more aligned with policymaker goals than alternative funding rules. This generates the first testable hypothesis of the paper: public universities respond to changes in expected funding, not the underlying structure of the rule.

The base model uses general assumptions to be widely applicable, but I show in Section 4.4 that it can be adapted to specific cases by using Ohio as a case study. For example, Ohio identifies "at-risk" students and gives additional funding for degrees completed by these students. While there exists a unique Ohio funding rule that realigns a university's enrollment problem with the state policymaker's (Proposition 3), I discuss why the existing Ohio funding rule likely fails to realign enrollment incentives. In particular, I argue that Ohio's "at-risk" funding likely under-incentivizes universities to enroll URM students. Finally, I find that changes in funding rules will disproportionately affect enrollment of URM students if and only if the university's private return to its marginally admitted non-URM student is relatively small compared to the private return of the marginally admitted URM student (Result 1). This result generates the second testable hypothesis: public universities disproportionately change enrollment of URM students in response to funding changes. Because Result 1 holds in both directions, if the testable hypothesis holds, then this implies that universities may be too risk averse over enrolling students with "risky" individual characteristics.

I test the two hypotheses using data from the Integrated Postsecondary Education Data System (IPEDS) and Common Core of Data (CCD), using Ohio's switch to PF as a case study. I give information on the data and descriptive statistics in Section 5. My empirical approach is difference-in-differences with synthetic controls, and the outcomes of interest are enrollment level and composition of enrollment among first-time, full-time students at Ohio public universities. I show that the switch to PF was concurrent with a short-run funding increase but long-run funding decrease. The main testable hypothesis therefore predicts an initial increase in enrollment level, followed by a long-run decrease. I also test for whether URM student enrollment was disproportionately affected by estimating changes in composition of first-time, full-time cohorts.

I present the main empirical results in Section 6. I find evidence of short-run enrollment increases, but that overall enrollment decreased by 1 percentage point in the long-run. This is consistent with the first testable hypothesis: public universities indeed responded more to level of funding than structure of the funding rule. The long-run effect is larger for Black students: enrollment level for Black students decreased by 2.8 percentage points. Furthermore, I find that the proportion of Black students decreased by 1.13 percentage points, with no evidence that universities systematically redistributed enrollment composition towards any specific group. This suggests that enrollment changes were disproportionately borne by Black applicants, which is consistent with the second testable hypothesis. An immediate policy implication is that public universities are likely under-incentivized to enroll URM students.

Because the empirical results are consistent with the theoretical predictions, I argue that this shows the theoretical framework can be be used to predict how funding changes may affect enrollment in different contexts and provide guidance to policymakers. I conclude in Section 7 by making recommendations on how the results from this paper may guide future policy. Two key policymaker takeaways are that policymakers should (1) create campus-level funding rules, not state-level funding rules, that take into account that universities have different institutional missions and therefore different enrollment preferences, and (2) carefully consider the incentive to enroll URM students to ensure that the funding rule does not unintentionally under-incentivize these students.

2 Literature Review

This paper contributes to three overarching strands of literature. First, this paper evaluates the use of pay for performance at public universities. Lazear (2000), van Thiel and Leeuw (2002), and Podgursky and Springer (2007) review the effects of pay for performance on the private sector, public sector, and in the K-12 educational sector respectively. Similar to this paper, Gupta (2021) examines the interplay between incentivizing improved outcomes while potentially introducing an unintentional incentive to inequitably increase selectivity in the context of healthcare. He finds that a program aimed at reducing hospital readmissions of Medicare patients was successful in its main goal, but that only half of the reduction is explained by the desired effect of increased quality of care, while the other half is explained by increased selectivity over returning parents.

Second, this paper is adjacent to research on how financing affects a university's operations that started from work by Brown, Dimmock, Kang, and Weisbenner (2014) on the importance of university endowments. This paper has similar comparative statics results as Bulman (2022), who finds that changes in an endowment particularly affects enrollment of URM students. I find evidence that changes in state funding can unintentionally target URM students, suggesting that this result is agnostic to the source of financing.

Third, this paper complements studies that study how PF affects *outcomes* at four-year institutions. The majority of the literature focuses on the intended goal of increasing degree completion; researchers thus far have not found conclusive evidence that PF increases degree completion. Like this paper, Hillman, Fryar, and Crespin-Trujillo (2018) and Ward and Ost (2021) study Ohio. Both papers find that while degree completion increased after PF, the increase was likely not caused by PF. Hillman, Tandberg, and Gross (2014), Kelchen (2018), Tandberg and Hillman (2014), and Rutherford and Rabovsky (2014) study states that have implemented smaller degrees of PF than Ohio and find similar results. Consistent with the theoretical results in this paper, other papers find that the effect switching to PF on outcomes may depend on institutional characteristics and policy-specific incentives. Boland (2020) finds that PF had no effect on degree completion at historically Black colleges and universities. Favero and Rutherford (2019) and Hagood (2019) find that PF may disproportionately increase funding at universities that already perform well and are less resource constrained. Similarly, Birdsall (2018) finds institutions that are less dependent on state funding increase graduation rates after switching to PF. Finally, Li (2020) finds that adding a premium on STEM degrees increases the number of degrees conferred in STEM fields.

It is less clear how PF affects college admissions. Kelchen (2018) finds that PF had no effect on enrollment. Gandara and Rutherford (2020) find that universities admit cohorts with higher standardized test scores after states switch to PF, and find some negative short-run effects on URM enrollment. Umbricht, Fernandez, and Ortagus (2017) study PF in Indiana and find some evidence that URM enrollment decreased, but these results are confounded by the fact that Indiana simultaneously implemented other higher education initiatives. Gandara and Rutherford (2018) show that adding premiums to degrees completed by URM students mitigate unintended negative enrollment effects. Ward and Ost (2021) find evidence that Hispanic enrollment decreased in the PF states, but enrollment is not the focus of their study. To my knowledge, I am the first to look at how PF affects long-run enrollment dynamics. This may explain why my empirical results differ from other studies: while I find a short-run increase in Black enrollment, which is consistent with existing results, I find a decline in enrollment of Black students begins 3 years after implementation.

For a full history of how PF has been used in the US, see Dougherty et al. (2016) and Dougherty and Natow (2015). Finally, Kaikkonen (2016) is a policy brief that discusses both theoretical and practical concerns in designing funding rules, using Washington state's Student Achievement Initiative as a case study for how funding rules are implemented. While this case study affected community colleges, many of the policymaker insights carry over to baccalaureate institutions; in particular, I draw upon discussion from Kaikkonen (2016) when considering gaps between the theoretical model and how PF was actually implemented in Ohio.

Table 1: Information on Ohio Public Universities

Institution Name	Enrollment	% In-State	6-Year Graduation %	Other Info
Bowling Green State University	13,853	88%	61%	
Central State University	5,406	55%	25%	Historically Black university
Cleveland State University	9,776	85%	49%	
Kent State University	20,418	80%	65%	
Miami University	16,864	62%	81%	
Ohio State University[1]	46,123	66%	88%	Public flagship, has hospital
Ohio University	18,113	88%	65%	
Shawnee State University	3,091	84%	35%	Open admissions
University of Akron[2]	11,323	92%	48%	Test optional admissions
University of Cincinnati	29,663	78%	72%	Has hospital
University of Toledo	11,965	74%	55%	Open admissions,[3] has hospital
Wright State University	6,938	95%	44%	Test optional admissions
Youngstown State University	8,822	76%	49%	Open and test optional admissions

Data are from the NCES College Navigator. Enrollment is the total undergraduate population as of Fall 2022. 6-year graduation rate is for the Fall 2016 cohort.
[1] I only include Ohio State University's main campus here, not its regional campuses.
[2] Test optional universities do not require applicants to disclose standardized test scores, but generally require a high school transcript.
[3] Open admissions universities automatically admit any students who meet a publicly posted minimum standardized test scores and/or high school GPA. Most also accept some students with scores below the threshold after a holistic review.

3 Policy Context in Ohio

Ohio has 14 public four-year universities. I exclude Northeast Ohio Medical University from analysis due to being a medical school and summarize information on the other 13 public universities in Table 1. The public universities are overseen by the Ohio Board of Regents, and all institutions belong to the Inter-University Council of Ohio. While the Board of Regents' role is regulatory, the Inter-University Council is a voluntary association among institutions.

Ohio's public universities primarily serve local students. In-state students make up >70% of most universities' student bodies, with the exceptions of Central State University, Ohio State University, and Miami University. Central State University likely attracts out-of-state students due to being a historically Black university, which are relatively rare in the Midwest. Ohio State University and Miami University may attract more out-of-state applicants due to being highly ranked public universities, according to measures such as the *U.S. News Best Colleges* (2023).

Next, I give an overview on the political history of higher education funding in Ohio. Like most states, Ohio historically appropriated funds to public universities based primarily on enrollment. In 2007, the Ohio 127th General Assembly (the legislative body of the state government) called upon the Chancellor of the Ohio Board of Regents to prepare a 10 year strategic plan for Ohio higher education. In 2008, the report was released; among other recommendations, the Chancellor outlined a plan to "convene a funding consultation to develop recommended changes in the state subsidy for higher education" (Fingerhut 2008). Later that year, Ohio's Inter-University Council published recommendations for a new state funding formula, *A Funding Formula for Ohio's Universities based on Outcome Goals* (2008). The main recommendations were:

1. To switch to a funding rule based on course completions, degree attainment, doctoral education/research funding, and medical education funding;

2. To weight funding based off of expected vs. actual graduation rates to incentivize "increasing the number of college graduates among those students less likely to graduate";

3. To set aside between 5%-10% of funding for institution-specific outcomes; and

4. To ensure stability and predictability in funding by applying a stop-loss in the first two years after switching to performance funding.

The 128th Ohio General Assembly enacted major changes to higher education funding in July 2009 under House Bill 1 (HB1) that were consistent with most of the Council's recommendations. At the time, the Ohio state Senate was majority Republican (21 Republicans, 12 Democrats) and the state House of Representatives was majority Democrat (46 Republicans, 53 Democrats). The bill was signed by then-Governor Ted Strickland (Democrat, 2007-2011). It bears mentioning that subsequent General Assemblies have had a Republican majority in both legislative bodies, and that the Ohio governorship has been held by a Republican since Strickland left office. The new funding rule was implemented in fiscal year (FY) 2010.

The next governor, John Kasich (Republican, 2011-2019) led an initiative to make several major changes to how funding rules were implemented. In 2012, Kasich formed the Ohio Higher Education Funding Commission. While the Inter-University Council had representatives from all of Ohio's public university, the Funding Commission's structure is different: presidents from universities and community colleges rotated in participation and chairmanship.[3] The 2014 Mid-Biennium Review legislation House Bill 484 (HB 484) enacted the recommended revisions.

I discuss next how four key characteristics of Ohio PF have evolved over time: the emphasis on degree completion, the stop-loss, funding for students identified as less likely to graduate ("at-risk" students), and funds for institution-specific goals. Broadly, the characteristics can be broken up into two time periods: as the funding rule existed under HB1 (the "original" version of performance funding) starting in FY 2010, and after FY 2014, when the changes spearheaded by Governor Kasich went into effect. I also summarize discussion below in Table 2.

Degree completions. Under HB1, degree completions determined 5% of state appropriations in FY 2010 and 10% in FY 2011. Subsequent legislation increased this to 20% in FYs 2012 and 2013. Initially, the weights on degree and course completions combined were to determine no more than 30% of funding by FY 2014 Pyle (2009). Under HB 484, the weight on degree completions sharply increased in FY 2014 to 50% (Carey 2014).

According to the Ohio *State Share of Instruction Handbook* (2022), state appropriations are currently based on course completions (30%) and degree completions (50%), with an additional 20% set aside for discretionary funding (e.g., a university could receive more money if it has a medical center). Out-of-state graduates not employed in Ohio do not count towards degree completion, while out-of-state graduates employed in-state count for half a completion.

Stop-loss. HB1 implemented stop-losses to prevent institutions from losing funding as a result of the switch to performance funding in FY 2010. The stop-loss was 99% in FY 2010 and 98% in FY 2011. Given the state appropriations a university received in FY 2009, the same university would receive no less than 99% of that amount in FY 2010. The initial stop-loss was particularly important for Shawnee State, Youngstown State, and Central State Universities—these institutions would have lost funding otherwise (Moltz 2009). This is expected, because Table 1 shows that these institutions have lower completion rates; however, these institutions also tend to serve local first-generation students (Moltz 2009).

The stop-loss was intended to make funding levels stable and predictable as the new rule came into effect.

[3]In 2012, the Funding Commission had representatives from 5 public universities (The Ohio State University [chair], Miami University, Wright State University, Shawnee State University, Ohio University) and 4 community colleges according to *Recommendations of the Ohio Higher Education Funding Commission* (2012). In 2014, it had representatives from 4 public universities (Ohio University [chair], The Ohio State University, Wright State University, Cleveland State University) and 3 community colleges according to *2014 Recommendations of the Ohio Higher Education Funding Commission* (2014).

However, it also served as a redistribution mechanism. In order to implement the stop-loss, the institutions that would have gained funding as a result of the new rule had their gains capped and redistributed to institutions that would have lost money in the absence of the stop-loss. For example, the public flagship Ohio State University had its funding gains capped in FY 2010 (Pyle 2009). HB 484 explicitly removed the stop-loss to end redistribution of funds, which advantages the institutions that already have high completion rates (Carey 2014).

Table 2: Summary of Funding Changes

	FYs 2010-2011 (Strickland)	FY 2014 (Kasich)
Emphasis on degree completion	FY 2010: 5% FY 2011: 10%	FY 2014: 50% Largest funding component
Stop-loss	FY 2010: 99% FY 2011: 98% Prevented institutions from losing virtually any funding	Removed
At-risk student subsidization	Institution-specific Redistributed funds from high-performing institutions to low-performing institutions Incentivized enrollment of at-risk students	Non-institution specific Ended redistribution of funds between institutions
Institution-specific funding	5% of total funding	Removed

At-risk students. While drafting HB1, the 128th General Assembly commissioned a study on how to identify students who are "at-risk" of academic failure based on identifiable demographic characteristics. The study also proposed at-risk "weights" (essentially subsidies) that provide additional funding for courses and degrees completed by at-risk students. Ohio policymakers explicitly included at-risk weights to incentivize enrollment of these students, given that they are less likely to complete courses and degrees. At-risk student weights are based on observed differences in historical graduation rates (Carey 2014). Weights were initially applied at the campus level, so universities with lower graduation rates that enrolled proportionally more at-risk students stood to benefit most from these at-risk premiums.

HB 484 began applying the at-risk weights at the student level without taking into account the campus that the student attends (Carey 2014). That is, Ohio State University receives the same additional funding for graduating a low-income student as every other university would, even though Ohio State University has the highest degree completion rate among Ohio public universities. This redistributed funding away from institutions that enroll more at-risk students.

According to Carey (2014), the current funding rule identifies two separate sets of at-risk student characteristics: one set for course completions and one set for degree completions. At-risk characteristics considered for course completions are financial and academic status. At-risk characteristics for degree completions additionally include age, race, and first generation status.

Institution-specific funding. HB1 set aside 5% of funding for "institution specific goals and metrics that will be negotiated through a process established by the Chancellor". Universities that failed to make sufficient progress on these goals would have up to 10% of the initial set-aside at risk for removal starting in FY 2011. Governor Kasich's 2014 policy changes completely removed institution-specific funding.

Table 3: Annual Tuition - Ohio State University

Academic Year	Dollar value	Percent change
2025-2026	$13,641	3%
2024-2025	$13,244	3%
2023-2024	$12,859	3%
2022-2023	$12,485	5%
2021-2022	$11,936	4%
2020-2021	$11,518	4%
2019-2020	$11,084	3%
2018-2019	$10,726	1%
2017-2018	$10,591	6%
2016-2017	$10,037	0%
2015-2016	$10,037	0%
2014-2015	$10,037	4%
2013-2014	$9,615	0%
2012-2013	$9,615	3%
2011-2012	$9,309	4%
2010-2011	$8,994	7%
2009-2010	$8,406	0%
2008-2009	$8,406	0%
2007-2009	$8,406	0%
2006-2009	$8,406	

Finally, I conclude with some additional contextual information. Tuition at Ohio's public universities was frozen starting in the 2006-2007 academic year and ending in the 2009-2010 academic year ("State higher ed chancellor praises tuition freeze" 2009). Even when the state doesn't mandate tuition bans, some public universities intentionally freeze tuition or keep tuition increases capped. For example, using the Internet Archive, I was able to track annual tuition for first-year in-state students at the public flagship, Ohio State University and have summarized the information in Table 3. Similarly, Kelchen, Ortagus, Rosinger, and Cassell (2022) find little evidence that students at four-year institutions in states that use performance funding increase educational debt, so any negative differences in funding are likely not being "absorbed" by increased student payments to institutions. Because tuition increases often controlled, institutions cannot rely on tuition increases to increase expected revenue, nor is there evidence that public universities are shifting payments onto students in other ways. This points to the importance of state appropriations as an incentive strong enough to affect enrollment.

4 Theoretical Framework

While Ohio is an important case study on the implementation of PF, policymakers in other states may rightfully have concerns about extrapolating from Ohio's experience because Ohio's public higher education system is not similar to many other states. For example, college applicants in Ohio enjoy a relatively large number of public regional universities. Also, as will be seen later in Section 5.1, Ohio enrolls a different demographic of students than other states. This motivates the usefulness of theoretical modeling: to provide a tool that can be adapted to more specific contexts and used to predict how changes in funding rules may affect enrollment.

In this section, I start by building a generalized model of college enrollment with a state funding rule. The goal of the model is to predict how different funding rule affect enrollment selectivity across different groups of students. I then adapt it to the specific case of Ohio in Section 4.4 to generate concrete, testable hypotheses on how the model predicts that switching to PF will affect enrollment, in the context of Ohio's specific policy characteristics. These testable hypotheses are then assessed in Section 6 to provide evidence that the model generates sensible predictions that state policymakers can use to guide future funding decisions.

4.1 General Model

There is one social planner (or state policymaker, SP), a finite set of universities indexed by k, and a continuum of in-state students. The SP and universities do not discount the future. In-state students differ across two dimensions: academic preparedness $a \in [0, 1]$ and group identity $g \in \{\text{M, URM}\}$ (majority and under-represented minority), so that the full type space is $\Theta = [0, 1] \times \{\text{M, URM}\}$. I focus on group identity as the at-risk characteristic of interest because it is relatively easy to identify in the data used in the empirical section of this paper, and because of its policy relevance. An arbitrary student i is described by her type $\theta_i = (a_i, g_i) \in \Theta$. Academic preparedness is a composite score that describes how strong a student's application is, and $a_i|g \sim Uniform[0, 1]$.[4]

There are three time periods, $t \in \{0, 1, 2\}$. At $t = 0$, the SP sets and credibly commits to a publicly observable funding rule. At $t = 1$, applications, admissions, and enrollment occur.[5] Students send applications to all universities they find acceptable, and universities admit all students they find acceptable.[6] All

[4] Assuming a uniform distribution simplifies notation and analysis, but I can let academic preparedness follow any arbitrary probability distribution over $[0, 1]$. For example, it may be more reasonable to assume ability is normally distributed, but then I must take into account the measure of students of every academic preparedness level. This does not qualitatively change theoretical results but adds modeling complexity, hence why I use the uniform distribution.

[5] In principle, the admissions process is likely capacity constrained by how many students a university is willing to eventually enroll. I discuss capacity constraints more in Appendix A.1; additionally imposing capacity constraints does not affect overarching results significantly.

[6] Admissions can be either sequential (with students applying first, then universities accepting students) or simultaneous. Outcomes are equivalent. That is, results are not sensitive to different assumptions on how admissions works.

accepted applicants then enroll at their favorite offer.

After enrollment, a student realizes one of three outcomes: she exits her university early during $t = 1$, exits late during $t = 2$, or graduates at the end of $t = 2$. This captures the spirit of existing PF rules: universities can receive funding for students who complete some education but fail to graduate. We can think of "early" exiters as students who don't complete general requirements, while "late" exiters have completed general requirements but not major-specific requirements. This simplification captures the broad strokes of college attrition: historically, about half of exits occur during the first year of enrollment.[7]

Denote the expected probability that i will exit before completing general requirements (i.e., before $t = 1$ ends) as a function of her type, $Early(a_i|g_i) : [0, 1] \to [0, 1]$, and the expected probability that she will exit before completing major requirements (i.e., before $t = 2$ ends) as $Late(a_i|g_i) : [0, 1] \to [0, 1]$.[8] The exit probabilities are known to the SP and university through prior student bodies. I impose two following assumptions on the exit functions.

Exit Assumption 1. *Let both exit functions be continuously differentiable and weakly decreasing in a_i.*

Exit Assumption 2. *Let $Early(a_i|URM) \geq Early(a_i|M)$ for all a_i, and the equivalent inequality holds for $Late()$.*

The first assumption implies that more academically prepared students are less likely to exit. The second assumption implies that a URM student with the same academic preparedness as a non-URM student is less likely to graduate in expectation.

4.2 General Social Planner's Enrollment Problem

First, consider the social planner's enrollment problem.[9] That is, if a social planner could directly admit students, which students would ultimately enroll where?

Suppose that student i enrolled at university k creates social returns to her enrollment. I assume there are two components to social returns: (1) the value of receiving education and (2) the value of degree completion. A student who completes only her general requirements generates social returns to her enrollment due to receiving education, but always less than if she had successfully completed the degree.[10]

[7] The *Yearly Success and Progress Rates: Fall 2015 Beginning Postsecondary Student Cohort* (2022) report finds that 78.9% of students in the Fall 2015 starting cohort at public four-year universities returned after their first year and 61.8% completed a degree at their starting institution, so 55% of attrition occurred during the first year.

[8] By construction, $Late(a|g) \geq Early(a|g)$ for all students, as a student who doesn't complete general requirements also does not complete major-specific requirements, but a student can exit before finishing her major but have completed her general requirements.

[9] I assume for now that the SP has complete information on students and university characteristics. This is an unrealistically strong assumption, but I impose it to show a naively constructed funding rule might fail to achieve its intended goals even in ideal circumstances.

[10] This is consistent with the fact that the median weekly earnings for people with some college education but no degree is

Denote the social returns to education if student i enrolls at university k and completes her general education during $t = 1$ as a function of her type,

$$Educ_k^{SP}(a_i|g_i) : [0,1] \to [0,1]$$

and similarly, denote the net social returns to i completing her major-specific requirements and earning a degree at k as

$$Degree_k^{SP}(a_i|g_i) : [0,1] \to [0,1].$$

These social returns returns satisfy the following two assumptions.

Social Returns Assumption 1. *Both functions are weakly increasing in a_i and continuously differentiable.*

Social Returns Assumption 2. $Educ_k^{SP}(a_i|URM) \geq Educ_k^{SP}(a_i|M)$ *and* $Degree_k^{SP}(a_i|URM) \geq Degree_k^{SP}(a_i|M)$ *for all a_i.*

The first assumption says that academic preparedness is positively correlated with future productivity. The second assumption rules out *only* the possibility that the SP systematically values the social returns to enrolling a URM student strictly less than non-URM enrollment. It is possible that the SP systematically values enrolling URM students *more* than non-URM students, but it is also possible that the SP is "race-blind" and does not consider group identity at all, only the academic preparedness of an applicant.[11]

Finally, let the expected operating costs of education per student in each time period $t \in \{1,2\}$ be $c_t \in [0,1]$, which is sunk by university k at the beginning of each time period.[12]

The expected social return to admitting i at k is therefore a function of her type,

$$\begin{aligned} EV_k^{SP}(a_i|g_i) = & -c_1 + (1 - Early(a_i|g_i))[Educ_k^{SP}(a_i|g_i) - c_2] \\ & + (1 - Late(a_i|g_i))Degree_k^{SP}(a_i|g_i). \end{aligned} \tag{1}$$

I assume that the per-period costs (c_1, c_2) are not "too large" to guarantee that a state policymaker will always want to enroll at least some students.[13]

higher than those with a high school diploma and lower than those with a bachelor's degree, according to a report released by the U.S. Bureau of Labor Statistics (*Education Pays* 2023). Although the returns to post-secondary education when the student does not graduate are not as well understood as returns to education conditional on graduation Lovenheim (2023) as well as Maurin and McNally (2008) find evidence that additional years of college education increase lifetime earnings given that the student graduates.

[11]I show in Appendix A.2 that a race-blind social planner is always more selective over URM students than non-URM students.

[12]By operating cost, I specifically mean the expected cost of running classes, which can be calculated based off of past years. This is part of Ohio's funding rule calculations. There may be variance in how much an arbitrary student costs outside of classes, but I allow this to be a part of the student type-dependent returns functions $Educ_k^{SP}()$ and $Degree_k^{SP}()$. This reduces the number of components that depend on student type, which is more tractable for later analysis.

[13]More rigorously, let the per-period costs (c_1, c_2) be such that for both groups, there exists some group-specific level of academic preparedness $\hat{a}_g \in (0,1)$ satisfying $EV_k^{SP}(a_g|g) < 0$ for all $a_i < \hat{a}_i$ and $EV_k^{SP}(a_g|g) \geq 0$ for all $a_i \geq \hat{a}_i$.

An enrollment decision can be characterized as a pair of cut-offs on level of academic preparedness, one for each group.[14] For an arbitrary cut-off a_g on enrollment of students in group g, the enrollment decision says that all students in group g with academic preparedness at least as high as a_g are admissible. Because the social returns to enrollment are weakly increasing in academic preparedness, following Azevedo and Leshno (2016), any such enrollment decision is *stable*. Stability implies that neither universities nor students have "justifiable regret" what institution a student enrolls at: students can do no better, because her more preferred universities would not accept her, and universities can do no better, because any additional students they would want to enroll have been admitted to a university that they prefer more.

The *socially efficient* enrollment decision maximizes the total expected social returns to education. With some minor abuse of notation, denote the socially efficient cut-off level of academic preparedness for students in group g when capacity constraints do not bind as a_g^{SP}.

Lemma 1. *Socially efficient cut-offs exist. The socially efficient cut-off for students in group g solves $EV_k^{SP}(a_g^{SP}|g) = 0$.*

The SP is willing to admit every student whose expected social returns to enrolling at k are weakly positive.[15] That said, it is likely that capacity constraints *do* bind for many public universities. I discuss how the socially efficient enrollment policy is derived under capacity constraints in Appendix A.1, but I note here that capacity constraints does not affect major theoretical results.

4.3 General University Enrollment Problem

In practice, state policymakers cannot directly enroll students: universities independently decide which students to admit in a way that best satisfies that university's institutional priorities. So next, we consider the university's admissions problem. A university k and social planner should agree on operating costs and expected exit functions, as these are based on historical information available to both. However, the university's private returns to educating a student will differ from the social returns. Denote k's private returns to educating i at $t = 1$ as $Educ_k(a_i|g_i) : [0, 1] \to [0, 1]$ and denote k's private returns to graduating i as $Degree_k(a_i|g_i) : [0, 1] \to [0, 1]$. I impose three assumptions on the private returns functions.

Private Returns Assumption 1. *$Educ_k(a_i|g_i)$ and $Degree_k(a_i|g_i)$ are weakly increasing in a_i.*

[14] This is in contrast to papers like Chade, Lewis, and Smith (2014), where the academic preparedness of an application is mapped to a probability of being admitted—in this model, students are either assigned a probability 0 or 1. These papers focus on highly selective schools that receive many applications from high-performing students and must consider other dimensions of the student to prevent admitting students over the school's capacity. At public universities, the cut-off rule is more realistic.

[15] The abuse of notation is because efficient cut-offs should be specific to k. However, I avoid comparing between cut-offs at different institutions, so I drop any indicator for k when describing the cut-offs in terms of academic preparedness a for ease of notation.

Private Returns Assumption 2. *$Educ_k(a_i|URM) \geq Educ_k(a_i|M)$ and $Degree_k(a_i|URM) \geq Degree_k(a_i|M)$ for all a_i.*

Private Returns Assumption 3. *$Educ_k^{SP}(a_i|g_i) > Educ_k(a_i|g_i)$ and $Degree_k^{SP}(a_i|g_i) > Degree_k(a_i|g_i)$.*

The first assumption implies that university expects higher returns from more academically prepared students. This is justifiable because these students are more likely to bring prestige to the institution and positively affect the university's ranking, external funding options, investment into research, etc. The second assumption says that, similar to the social planner, the university doesn't accrue strictly lower returns from educating and graduating URM students. It is possible for universities to systematically value URM education/graduation higher, but it is also possible that universities are race-blind.

The final assumption implies that *social returns* to enrolling a student are always strictly higher than the private returns *to the university*. As such, in the absence of additional enrollment incentives, the university would always enroll strictly fewer students than the social planner. This captures the fact that state appropriations are intended to pay for the enrollment of students whom universities could not afford to enroll in the absence of state funding. There are two reasons that justify this assumption. First, a university may take into account non-operating costs that a policymaker does not explicitly consider in the cost parameters c_1, c_2 (e.g., outreach/retention programs, health services, community centers, or adding resources to admissions offices to increase the quality of the admissions process indirectly). Second, a policymaker may account for positive externalities to higher education in the social returns to higher education that the university does not. Also note that if, for some reason, a social planner valued enrollment systematically *less* than universities, then the policymaker wouldn't appropriate any funds in the first place. The fact that state appropriations are paid at all is therefore indicative of the fact that state policymakers would like universities to enroll more students.

Put altogether, the expected private return to university k for admitting student i of type (a_i, g_i) in the absence of additional funding is

$$\begin{aligned} EV_k(a_i|g_i) = &- c_1 + (1 - Early(a_i|g_i))[Educ_k(a_i|g_i) - c_2] \\ &+ (1 - Late(a_i|r_i))Degree_k(a_i|g_i). \end{aligned} \tag{2}$$

Optimal cut-offs for the university—that is, cut-offs that maximize the private returns to enrollment—exist.

Lemma 2. *The university's optimal cut-off for students in group g solves $EV_k^{SP}(a_g^*|g) = 0$.*

It immediately follows from the third private returns assumption that the university is more selective than the state policymaker if it receives no outside funding. So while a state policymaker cannot directly affect admissions, she can change the university's enrollment incentives via a funding rule. Since the policymaker might give additional funding for URM students, let there be up to four funding components (2 time periods times two groups of students). Denote $p_{t,g} \in \mathbb{R}_+$ as funding given at time $t \in \{1, 2\}$ for a student in group g provided she meets the rule's criterion. I define two classes of funding rules:

1. Enrollment funding (EF): Funds are awarded for every student at the beginning of a time period, before outcomes are realized. If an EF rule awards a university $p_{t,g}$ for every student in group g enrolled at time t, then the university's expected private return for admitting a student becomes

$$\begin{aligned} EV_k(a_i|g_i) = &- c_1 + p_{1,g} + (1 - Early(a_i|g_i))[Educ_k(a_i|g_i) - c_2 + p_{2,g}] \\ &+ (1 - Late(a_i|g_i))Degree_k(a_i|g_i). \end{aligned} \tag{3}$$

2. Performance funding (PF): Funds are awarded conditional on courses completed at the end of $t = 1$ and degrees completed at the end of $t = 2$. If a PF rule awards a university $p_{t,g}$ for every student in group g who completes time period t, then the university's expected private return for admitting a student becomes

$$\begin{aligned} EV_k(a_i|g_i) = &- c_1 + (1 - Early(a_i|g_i))[Educ_k(a_i|g_i) - c_2 + p_{1,g}] \\ &+ (1 - Late(a_i|g_i))[Degree_k(a_i|g_i) + p_{2,g}]. \end{aligned} \tag{4}$$

Call a funding rule "socially efficient" or "socially optimal" if implementing the funding rule realigns the university's optimal cut-offs with the socially efficient cut-offs as derived from Lemma 1 and inefficient otherwise. Socially efficient EF and PF rules exist.

Theorem 1. *The following rules implement the socially efficient cut-offs.*

1. *Pure EF:*

$$\begin{aligned} p^{EF}_{1,g} &= (1 - Early(a^{SP}_g|g))[Educ^{SP}_k(a^{SP}_g|g) - Educ_k(a^{SP}_g|g)] \\ p^{EF}_{2,g} &= \frac{(1 - Late(a^{SP}_g|g))}{(1 - Early(a^{SP}_g|g))}[Degree^{SP}_k(a^{SP}_g|g) - Degree_k(a^{SP}_g|g)]. \end{aligned}$$

2. *Pure PF:*

$$\begin{aligned} p^{PF}_{1,g} &= Educ^{SP}_k(a^{SP}_g|g) - Educ_k(a^{SP}_g|g) \\ p^{PF}_{2,g} &= Degree^{SP}_k(a^{SP}_g|g) - Degree_k(a^{SP}_g|g). \end{aligned}$$

Proof. The proposed funding rules pay the university k the difference between the social returns and the private returns to enrolling a student at the socially efficient cut-off for each group and time period. That is, these funding rules make the student at the socially efficient cut-off for each group just admissible to the university. Because $EV_k(a|g)$ is weakly increasing in academic preparedness, any student in group g with academic preparedness $a \geq a_g^{SP}$ has weakly larger private returns to k than the student at the group cut-off given the proposed funding rules, so is also admissible. □

A key takeaway from Theorem 1 is that switching from EF to PF *does not* inherently move universities closer to socially efficient enrollment because both can incentivize universities to pick the socially efficient cut-offs. Moreover, the *pure* EF and PF rules are not the only funding rules that implement the socially efficient cut-offs. As long as the funding rule awards the university the efficient amount of funding *in expectation*, the SP can spread payments across the first and second time periods.

Corollary 1. *Socially efficient EF and PF rules are not unique.*

Proof. Take the socially optimal PF rule from Theorem 1. Let $f, b \in [0, 1]$ (for "front-loaded" and "back-loaded" funding respectively). Any performance funding rule that satisfies

$$p_{1,g} = f \cdot p_{1,g}^{EF} + \frac{(1 - Late(a_g^{SP}|g))}{(1 - Early(a_g^{SP}|g))} \cdot b \cdot p_{2,g}^{EF}$$
$$p_{2,g} = (1 - f) \cdot \frac{(1 - Early(a_g^{SP}|g))}{(1 - Late(a_g^{SP}|g))} \cdot p_{1,g}^{EF} + (1 - b) \cdot p_{2,g}^{EF}$$

also implements the socially optimal admissions cut-offs. The same is true for an appropriately constructed linear combination of the pure PF rule. □

In the generalized model, comparative statics are straightforward. Whenever an arbitrary payment component $p_{t,g}$ increases for students in group g in either time period $t \in \{1, 2\}$, the private returns to the university of enrolling a student in group g increases, so enrollment of students in group g increases. This is the same regardless of the structure/timing of the funding rule.

4.4 Modeling Performance Funding in Ohio

I now show how the general model can be adapted to analyze and predict how changes in funding rules will affect more specific environments using Ohio as a case study.

An immediate implication from Theorem 1 is that efficient funding rules are implemented at the *institution level*. Institutional missions differ; for example, regional universities are often more oriented towards enrolling local students, while a public flagship likely has more prestige concerns. However, Ohio implements its funding rule at the state level. Given the heterogeneity between Ohio public universities, a statewide funding rule is therefore efficient for at most one institution. This was a key takeaway from the case study on PF that Kaikkonen (2016) describes: universities must be treated as stakeholders in the funding system they participate in, or else they may not respond to incentives as intended. Looking at Table 2, PF was initially implemented at the institution level and moved in FY 2014 to the state level. While this cannot be empirically tested because we don't know what the "socially efficient" enrollment level is, this does imply that the post-FY 2014 implementation of PF is likely to be less efficient than the earlier iteration because it removes 2 campus-level funding determinants.

Putting aside the level of implementation, the model must be adjusted to account for the fact that Ohio imposes additional constraints on its PF rule that create restrictions across time and between groups:

1. Course completions are treated the same between groups, since URM status is not an at-risk characteristic for course completions. Therefore, $p_{1,URM} = p_{1,M}$.

2. Ohio awards different levels of funding to URM vs. non-URM degree completions. This enters in as a subsidy for URM degree completions. Denote this subsidy as $\psi \in \mathbb{R}_+$, so that $p_{2,URM} = (1+\psi)p_{2,M}$.

3. Ohio awards 30% of funding based on course completions and 50% of funding based on degree completions. To simplify, assume that discretionary funding enters the university's enrollment problem as a lump sum, so it doesn't affect the university's cut-off choices. Then we can think of funding as follows: the SP is willing to fund up to $p \in \mathbb{R}_{++}$ per enrolled student who eventually completes a degree. A fraction $\phi \in [0, 1]$ of total funding p is paid at the end of $t = 1$ conditional on course completion and the rest is paid conditional on degree completion. That is,

$$\begin{aligned} p_{1,M} &= (1-\phi)p, \\ p_{2,M} &= \phi p. \end{aligned}$$

Summarizing these restrictions, I define the class of "Ohio PF rules" as $\rho^{OH} = (p, \phi, \psi)$ such that:

1. $(1-\phi)p$ is paid for every student still enrolled at the end of $t = 1$,

2. ϕp is paid for every student who graduates at the end of $t = 2$, and

3. An additional amount $\phi \cdot \psi p$ is paid for every URM student who graduates at the end of $t = 2$.

Even with these restrictions, there theoretically exists an institution-level Ohio PF rule that induces universities to choose the socially efficient enrollment cut-offs.

Proposition 1. *A socially efficient Ohio PF rule exists. It is unique.*

The full proof is in Appendix A.3. The intuition is similar to Corollary 1: the SP can front-load or back-load funding so long as the funding rule gives the efficient amount of funding in expectation. It is unique because of the restrictions on the class of Ohio PF rules. To find the socially efficient Ohio PF rule, the social planner first chooses the expected total funding amount p and the relative weight on funding given in the first period ϕ to induce the socially efficient cut-off for students in group M. Then ψ is used as a back-loaded compensation to subsidize enrollment for URM students that accounts for differences in social returns in *both* time periods, but that is only paid at the end at $t = 2$. Since the university anticipates receiving the socially efficient payment *in expectation*, this realigns the university's enrollment decision with the social planner's.

In Ohio, the actual subsidy for URM degree completions is based *only* on "the decreased likelihood of students graduating based on their risk category" (*State Share of Instruction Handbook* 2022). This implies that Ohio's URM subsidy is

$$\psi^{OH} = Late(a^{SP}_{URM}|URM) - Late(a^{SP}_{M}|M) > 0, \tag{5}$$

where the inequality is imposed because the URM subsidy used in practice is strictly positive.[16] This is a restrictive condition that likely causes universities to be inefficiently over-selective with URM students, which I discuss in more Appendix A.3, Corollary 2.

It bears noting that if the URM subsidy is inefficient, it is *not* because of differences in how state policymakers vs. universities evaluate *students* based on group identity. That is, the socially efficient subsidy doesn't depend on either the social planner's or state policymaker's "tastes" for enrolling URM students. The problem with Equation 5 would persist even if the social planner and university were both completely race-blind. The reason Equation 5 is likely inefficient is because a socially efficient subsidy should realign the social and private returns to enrollment. Fundamentally, social and private returns differ because either (i) the social planner accounts for additional benefits like the positive externalities of education, (ii) the university accounts for additional costs of enrollment, or (iii) some combination of both. Equation 5 does something fundamentally different: it compensates for the difference in URM vs. non-URM graduation rates. Except in unusual knife-edge cases, it is unlikely that these two gaps would exactly coincide.

[16] According to *State Share of Instruction Handbook* (2022), this is the closest to how Ohio state policymakers calculate ψ^{OH} in practice. However, analysis would not change much if ψ^{OH} were instead an arbitrary function of the difference in graduation rates $f(Late(a^{SP}_{URM}|URM) - Late(a^{SP}_{M}|M)) : [0,1] \to \mathbb{R}_+$ (e.g., taking a proportion of the difference); simply substitute $Late(a^{SP}_{URM}|URM) - Late(a^{SP}_{M}|M)$ with $f()$ in Corollary 2.

4.5 Comparative Statics in the Ohio Model

I conclude the theoretical section of this paper by characterizing how small changes in an Ohio PF rule may affect enrollment to generate testable hypotheses. For the full comparative statics results, see Proposition 4 in Appendix A.4, where there is also additional discussion of theoretical results. The broad strokes of the results follow:

Result 1. *Under non-restrictive conditions on a university's private returns to enrollment, the university:*

1. *Generally becomes more selective as the overall funding level decreases,*
2. *Becomes more selective over students in the URM group as the URM subsidy decreases.*
3. *Becomes more selective over students in the non-URM group as the funding rule shifts away from enrollment funding and towards performance funding. Except for extremely large URM subsidizes, which distort the incentive to enroll students in the URM group, the same is true for students in the URM group.*

The first empirically testable hypothesis follows from this result.

Testable Hypothesis 1. *If changes in funding indirectly causes a decrease (increase) in funding, then overall level of enrollment also decreases (increases).*

Recall that an argument against using PF is that it targets enrollment of URM students. It is therefore of interest to understand if funding changes disproportionately affect URM enrollment. For the full result, see Proposition 5 in Appendix A.4. The broad strokes of the result follow:

Result 2. *Under non-restrictive conditions on a university's private returns to enrollment, a change in level of funding causes a disproportionate change in enrollment of URM students (relative to that of non-URM students) if and only if either (or both) of the following conditions hold:*

1. *The subsidy for degrees completed by URM students is very large, and/or*
2. *The university's private return to the marginally admitted non-URM student is relatively small compared to the private return to the marginally admitted URM student.*

Note that this result *doesn't* predict whether URM students are disproportionately over- or under-enrolled compared to non-URM students—only whether a funding rule would cause the incentives for URM enrollment to be systematically different from the incentives for non-URM enrollment in an undesirable way. The intuition behind this result is that there are two overarching factors that can distort the incentive to enroll

URM students relative to non-URM students. The first is straightforward: if the URM "subsidy" is too large, it distorts the incentive towards over-enrolling URM students. The second reason is that the *private return* to the university from enrolling the marginal non-URM student is small relative to the marginal URM student. If the marginal gains from enrolling the last URM student are higher, then more URM students would be enrolled under any socially efficient funding rule. Such a university, in a sense, is risk averse: it prefers to enroll the "surer bet" of the marginally acceptable non-URM student, likely because her probability of completion is higher.

Result 2 holds in both directions. Also, from Table 2, note that the subsidy paid for URM completions was decreasing over time. Based off of this and analysis from Proposition 3, it is unlikely that the subsidy for degrees completed by URM students is large enough to disproportionately distort the incentive to enroll URM enrollments. Therefore, the more interesting condition is the second one and the following testable hypothesis is of interest:

Testable Hypothesis 2. *There is a disproportionate change in the enrollment of URM students after Ohio changed its funding rule.*

If Testable Hypothesis 2 holds, then Result 2 implies that universities would get higher private returns from admitting more URM students. This is important information to policymakers, as it suggests that universities are under-incentivized to enroll URM students who could generate larger returns, both private and social, to their enrollment than the marginally admitted non-URM students.

5 Data and Empirical Approach

The theory generates two empirical questions to test. First, does enrollment respond to the timing of a switch to PF or to changes in funding level? Second, do enrollment changes disproportionately affect URM students? To answer these questions, I use data from the Integrated Postsecondary Education Data System (IPEDS) and Common Core of Data (CCD). IPEDS contains annual institution-level data on U.S. universities and colleges that receive Title IV funding. I use enrollment data from 1995-2020 on post-secondary institutions that primarily award baccalaureate degrees and were public four-year institutions for the entire time period. The CCD contains annual data on the U.S. public elementary and secondary educational systems. I use education agency membership data from 2000-2020.

5.1 Descriptive Statistics

Using IPEDSD data, I show comparative trends in first-time, full-time enrollment at Ohio's public universities using all continental states except Tennessee ("non-PF states") as the comparison group. [17] Also, overall summary statistics on mean enrollment from 1995-2020 can be found in Table 4. It is immediately apparent that Ohio public universities enroll more students than other states on average, except for Hispanic students. This is likely due to demographic differences: Ohio has a relatively small Hispanic population compared to the national average, and Table 1 shows that Ohio public universities primarily serve in-state students.

To inspect enrollment composition, I also look at the percentage of students in a given demographic over time. As is reflected in Table 4, Ohio public universities enroll proportionately fewer Hispanic students and proportionately more white students than the average non-PF state. However, visual inspection suggests that parallel trends roughly hold for these two groups before 2009. However, the proportion of Black students is noticeably different in Ohio compared to the average non-PF state across the entire time horizon. Among non-PF states, proportion of Black students enrolled was constant between 10%-12%. In Ohio, the proportion of Black students grew from just below 10% to almost 14% between 1995-2011 before sharply dropping back to below 10% between 2012-2017, and dipped below 10% trending towards 8% between 2018-2020.

Table 4: Mean Enrollment

	Non-PF States	Ohio
Total Enrollment	16,781.47	38,410.65
	(15,260.24)	(3,539.60)
AAPI Enrollment	1,255.72	1,014.38
	(3,048.22)	(263.16)
Black Enrollment	1,868.77	4,101.73
	(2,131.39)	(790.17)
Hispanic Enrollment	1,873.47	1,098.38
	(4,597.56)	(503.07)
White Enrollment	10,454.28	29,617.92
	(7,133.27)	(1,326.09)

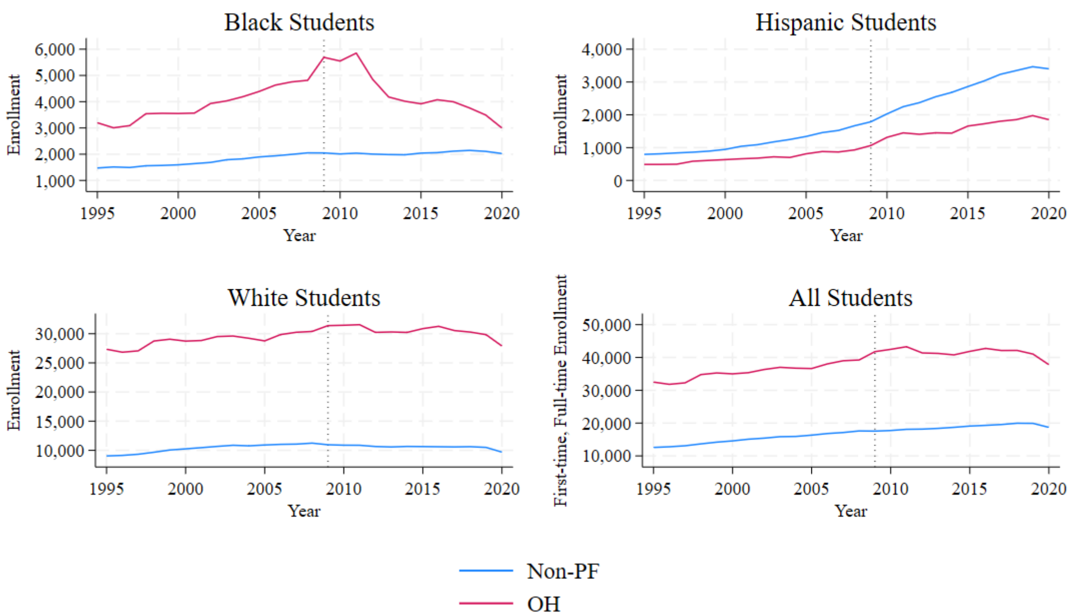
Next, I show trends in state funding of higher education. Panel (a) of Figure 2 shows total state appropriations in Ohio vs. non-PF states using both nominal dollar values (solid lines), as well as an inflation-adjusted value indexed to 2023 prices (dotted line).[18]

[17] I don't include Tennessee since Tennessee is another state that uses large amounts of PF to appropriate state funds to public universities. I discuss the case of Tennessee separately in Appendix E.

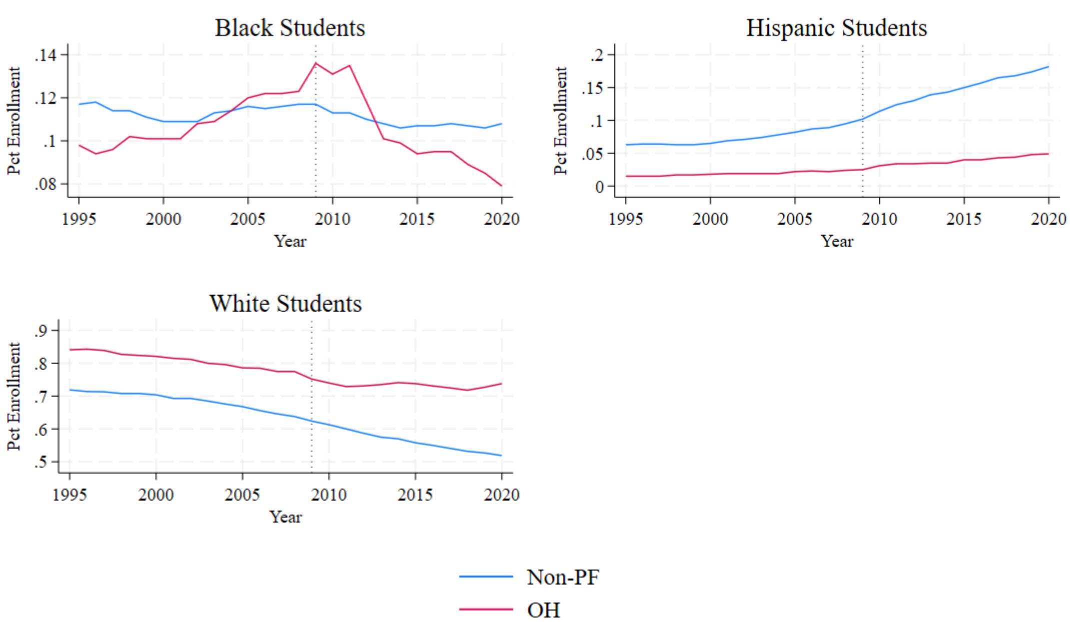
[18] I use the "Higher Education Cost Adjustment" (HECA), which is intended to account for inflation in the institutional costs of providing higher education. Alternate measures of inflation include the standard Consumer Price Index (CPI) and the Higher Education Price Index (HEPI). Following the *Technical Paper A: Inflation Adjustment* (2017), the standard CPI is not representative of the cost outlays that universities face, and the HEPI is partially endogenously determined (Blom, Washington, Rainer, Chien, and Kelchen 2020). Hence, my preferred inflation index is the HECA, but trends do not change substantially using the CPI or HEPI instead.

Figure 1: Enrollment Over Time

(a) Total First-time, Full-time Enrollment



(b) Percentage of First-time, Full-time Enrollment



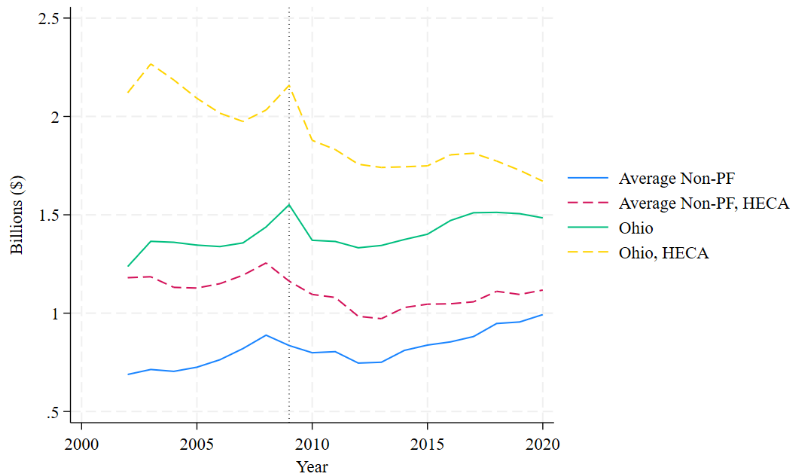
Trends in state appropriations were similar in Ohio compared to the non-PF states before 2008, with states modestly increasing appropriations over time. There is divergence from 2008 onward. First, the average non-PF state increased appropriations in 2008—likely in response to the Great Recession—but average appropriations returned to the pre-2008 level and trajectory in the year following. On the other hand, Ohio continued increasing state appropriations between 2009-2010, which is because PF was implemented and the initial stop-losses prevented funding losses. After the one-time spike between 2008-2009, though, the inflation-adjusted trend in state appropriations in Ohio trended very similarly to pre-implementation years: decreasing modestly over time. In contrast, inflation-adjusted average state appropriations in the non-PF states remained fairly constant from 2010 onward.

Panel (a) of Figure 2 shows that Ohio has higher levels of funding than other states on average, but this is likely driven by the fact that Ohio universities also enroll more students on average, as can be seen in Table 4. To account for this, I also show nominal and inflation-adjusted state appropriations per full-time enrolled (FTE) student in panel (b) of Figure 2. While funding per FTE student in the non-PF states was comparable to funding per FTE student in Ohio before 2009, there is notable divergence after 2010. Between 2012-2020, Ohio appropriated between \$1,000-\$1,500 less per FTE student than the average non-PF state.

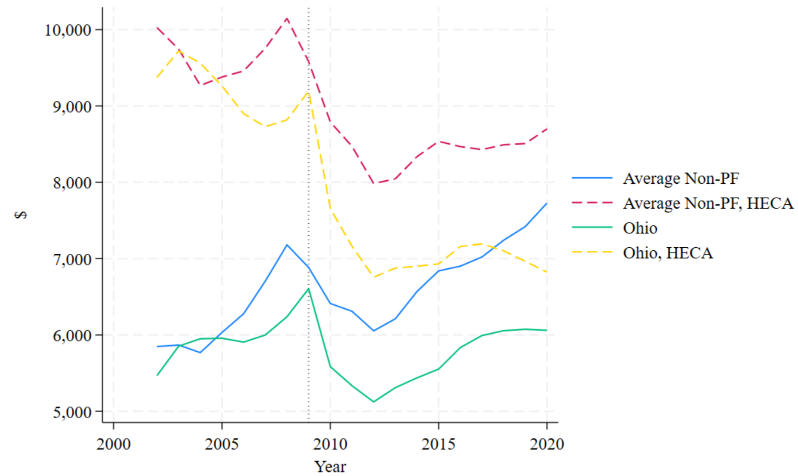
I also show the HECA-adjusted value of Ohio state appropriations at the institution level in panels (c) and (d) of Figure 2. The order that institutions appear on the legend correspond to their *U.S. News Best Colleges* (2023) ranking. Except for Central State University, all institutions experienced a one-time funding increase between 2008-2009. Afterwards, there is considerable heterogeneity in funding trends. Some institutions, like the flagship Ohio State University, experienced a long-run decline in state appropriations. In fact, the three highest-ranking institutions (Ohio State University, Miami University, the University of Cincinnati) received lower state appropriations after the switch to PF. Some regional universities, such as Bowling Green State University and Youngstown State University, eventually return to post-implementation levels of funding. Finally, institutions including the University of Akron and the University of Toledo receive more funding after PF implementation in the long-run. This suggests considerable heterogeneity in how PF affected each institution.

Figure 2: State Appropriations over Time

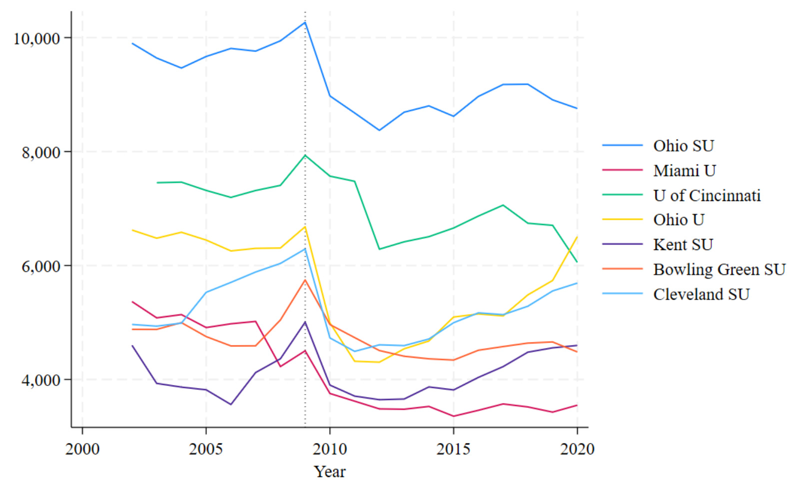
(a) Total Appropriations



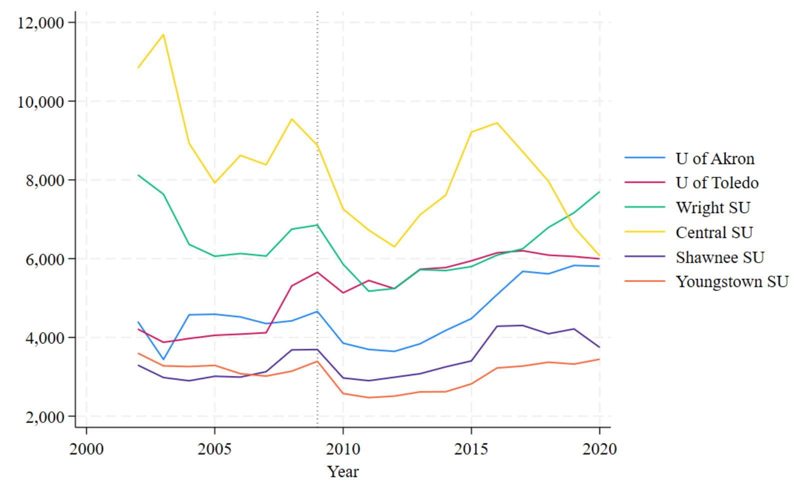
(b) Appropriations per FTE Student



(c) Appropriations per FTE, Ohio Institutions [1]



(d) Appropriations per FTE, Ohio Institutions [2]



5.2 Outcomes of interest

My state-level outcomes of interest are: (1) enrollment level, measured the number of students enrolled at any public university divided by the number of 12th grade students in a given demographic group (measure of in-state "enrollment level"); and (2) proportion of Black enrollment, measured as the number of first-time, full-time Black students divided by the number of all first-time, full-time students enrolled (proportion of Black enrollment). I measure all outcomes at the state level.[19] I use percentages to scale outcomes relative to the size of potential/overall enrollment, as Figure 1 shows that Ohio has higher enrollment numbers than non-PF states on average.

Enrollment level. An increase in enrollment level is equivalent to a decrease in selectivity. Analysis on enrollment level is to test Hypothesis 1, which said that universities respond to changes in *expected funding*, not the switch to performance funding. Figure 2 shows that state appropriations initially increased when Ohio switched to performance funding, but decreased in the long-run. Also, Table 2 showed that in FY 2014, Ohio sharply increased the emphasis on degree completion and removed various funding components that would generally have benefited universities with lower graduation rates. Table 1 showed that while the public universities with the highest completion rates also tend to be large institutions (Ohio State University, Miami University, University of Cincinnati), Ohio has a large number of regional universities with lower completion rates. Because larger, higher-performing universities are more likely to be capacity constrained in the first place, I argue that the FY 2014 changes likely had the largest impact on enrollment via these regional universities with lower completion rates. To summarize, Hypothesis 1 predicts that the initial switch to performance funding in FY 2009 should increase enrollment level (decrease selectivity), but that enrollment level should decrease by FY 2014.

Ideally, I would directly calculate enrollment level as the proportion of *in-state applicants* who apply to at least one public university in-state and eventually enroll, but this is not possible with the data available. I discuss briefly the gaps between my ideal measure and the measure that I actually use. There are three data limitations.

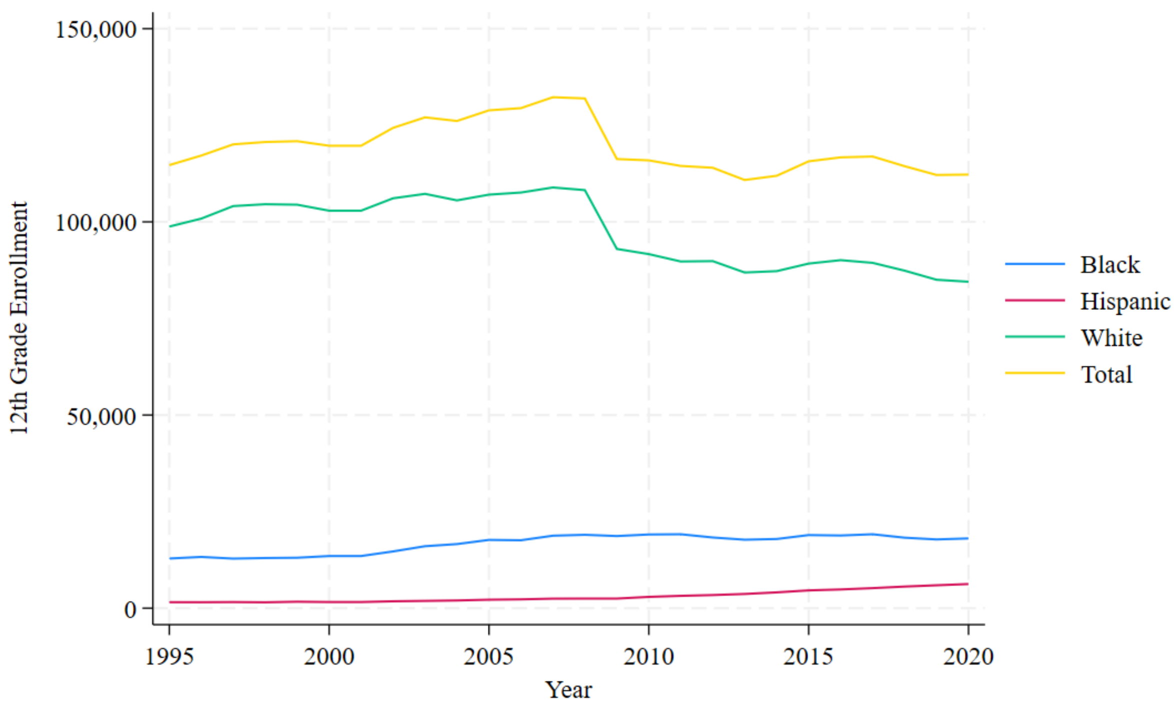
First, I cannot group enrollment data by demographic group and locale simultaneously using IPEDS data, so I group only by demographic group. This can potentially be an issue because universities may substitute between in-state and out-of-state students. For example, Bound, Braga, Khanna, and Turner (2020) show universities enroll more international students when facing funding cuts and Arcidiacono, Kinsler, and

[19] Outcomes are more volatile at the institution level, leading to institution level synthetic controls that are not well-matched. I aggregate to the state level to smooth out this volatility. Institution-level volatility is expected because public universities within Ohio also compete with each other for students, which could lead to enrollment between institutions tending to correlate with each other. By aggregating to the state level, the analysis is agnostic about *where* in-state students enroll and focuses on whether enrollment occurs at all.

Ransom (2023) show that universities are more selective over out-of-state students. Therefore, it is likely that public universities imperfectly substitute between non-local and local students. However, as seen in Table 1, public universities in Ohio predominantly enroll in-state students. In-state enrollment changes should dominate the *overall* effect on enrollment, although including out-of-state students in the measure of enrollment level attenuates the estimate towards 0.

Second, I use the population of in-state 12th grade students as the pool of potential applicants, but not all high school students apply to public four-year universities.[20] The denominator therefore overestimates the pool of potential college applications and also attenuates estimates towards zero.

Figure 3: 12th Grade Enrollment



Third, there is an accounting issue with the CCD data. As seen in Figure 3, Ohio *appears* to experience a one-time drop in 12th grade enrollment that exactly coincides with the time of PF implementation, from 134,522 in 2008 to 118,872 in 2009—a difference of 15,650 students (or an 11.6% drop).[21] After discussion with the Ohio Department of Education, I was able to confirm that this is likely due to an accounting change at the state level, as there was a business rule change on how to count students that were previously counted more than once in the data submission. Therefore, this does not indicate a *true* decrease in 12th grade

[20]The CCD has limited public data on the number of high school completions, and IPEDS does not have detailed applicant data. Also, using the total number of applications received at the state level is a worse candidate to use as the denominator—potential students may send multiple applications.

[21]This decrease also affects 11th grade enrollment in Ohio between 2008-2009 with roughly the same magnitude. However, lower grades are not affected. This one-time decrease does not appear in other states.

enrollment, but leaving the data untreated means that there will mechanically be an increase in enrollment level exactly at the time of PF implementation (FY 2010). To treat the data, I remove the one-time drop in number of white 12th grade students by adding back the approximate size of the decrease from 2009 and onward.[22]

While acknowledging that data limitations exist, the measure of enrollment level is still useful for analysis. The first two issues attenuate estimates towards 0, which means that enrollment level estimates are a lower bound. I have also taken steps to correct the data for inaccuracies in 12th grade enrollment. Also, it is reassuring that graphical inspection of Figure 3 shows that the populations of Black and Hispanic 12th grade students in Ohio are steady across time in the CCD data, suggesting that these sub-populations were not affected by the accounting change. So, enrollment level estimates for Black and Hispanic students can be meaningfully interpreted without modification.

Finally, the Ohio funding rule continued to evolve after the initial switch. In particular, Table 2 summarized changes in *how* Ohio implemented performance funding between FY 2010-2012 and after FY 2014. While the accounting change means that analysis right around FYs 2009-2010 should be taken with a grain of salt, analysis on the *dynamics* of performance funding post-implementation are not affected.

Proportion of Black enrollment. This measure captures changes in enrollment composition. A decrease in proportion of Black enrollment indicates that this demographic was disproportionately affected by changes in enrollment. Although Hispanic students are also under-represented minorities, I estimate proportions of Black and Hispanic enrollment separately, as Black, Cortes, and Lincove (2020) show that there are differences in application behavior and applicant readiness across Black and Hispanic students. Hispanic students have higher average college readiness than comparable Black students, but have a lower propensity to apply. As such, universities may perceive the academic readiness of Hispanic applicants differently from Black applicants of similar academic preparedness.

Analysis on proportion of Black enrollment is to test Hypothesis 2, which said that Ohio's switch to performance funding disproportionately affected URM students. If the hypothesis holds, then Result 2 tells us that universities likely under-enroll URM students. Graphical inspection of Figure 1 shows that enrollment of Black students in Ohio changed sharply several times between 2009-2014; however, it is not clear without a comparison group whether those changes are significant.

[22] Results do not change substantially by subtracting the approximate size of the decrease before 2009.

5.3 Synthetic Controls

My empirical approach is difference-in-differences with synthetic controls following Abadie, Diamond, and Hainmueller (2010).[23] The treatment group contains only Ohio. When using synthetic controls, the research has leeway in selecting both the matching variables used to create the synthetic comparison as well as the candidate pool to draw from. To avoid specification searching, I perform robustness checks to show that results are not sensitive to changes in either the candidate pool or the matching variables in Appendices B and C respectively.

My preferred matching approach is to match on all pre-treatment outcomes. I follow recommendations by Ferman, Pinto, and Possebom (2020) and also show results using alternate matching specifications in Appendix C. The main results hold under alternate specifications, though inference generally differs, as is expected. Because I use two datasets, the pre-treatment period begins later for enrollment level (2000) than it does for proportion of enrollment (1995).

My preferred candidate pool for the synthetic controls includes 36 states for enrollment level and 37 states for proportion of Black enrollment. I drop Alaska, Hawaii, Texas, and states in the Far West and New England regions.[24] For enrollment level, I additionally drop Idaho due to missing data on 12th grade students. However, I show in Appendix B that my results are robust to changes in the candidate pool.

I am selective with states in my preferred candidate pool for two reasons. First, adding more states in tends to result in synthetic controls that assign more weights than pre-treatment outcomes. Second, I believe that the states I drop from the candidate pool are poor comparisons to Ohio. I drop California and Texas due to their top percentage admissions policies, which create different enrollment restrictions for public universities. I argue that Alaska, Hawaii, the Far West, and Southwest states are not comparable to Ohio due to differences in underlying demographics and public university characteristics. These states (other than California, which is already dropped) have a small number of public four-year universities; for example, Wyoming has *only one* public university. In Ohio, in-state students face a variety of options, and the 13 Ohio public universities have significant heterogeneity over institutional mission and location. I also drop the New England states because of the small number of in-state options, and because geographic proximity in that region as well as a concentration of comparable private institutions may cause enrollment behavior in the New England states to be systematically different than in Ohio. I always include the states that I

[23]Traditional difference-in-differences requires the researcher to assume that the treatment group (Ohio) would have shared common trends with the control group, had PF not been implemented in Ohio. One way of assessing the common trends assumption is to assess how closely trends in the treatment and control groups correlated *before* the treatment. But Figure 1 shows that enrollment of Black students in the PF and non-PF states did not trend similarly before 2009. Focusing on specific control groups like Midwestern states does not improve common pre-trends. Hence, my preferred specification uses synthetic controls. Although not presented in this version of the paper, I find comparable results using difference-in-differences with two-way fixed effects using non-PF states as the control.

[24]I use the IPEDS classifications. Far West states include Alaska, California, Hawaii, Nevada, Oregon, and Washington. New England states include Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont

believe are the closest peer comparisons to Ohio: the Midwest and Mid-Atlantic states. In particular, some states that are both demographically similar to Ohio *and* have a similar distribution of public universities are therefore should definitely be included are Michigan, Illinois, Pennsylvania, and New York.

I also perform the same synthetic control analyses for private, not-for-profit four-year universities in Appendix D as a placebo test: private universities do not receive state funding and should not be affected by state funding changes, but are otherwise exposed to the same conditions as nearby public universities. I do not find any evidence that enrollment at private universities responded to changes in state funding, which supports that the main results indeed capture the causal effects of switching to PF.

6 Results

6.1 Enrollment Level

The enrollment level results appear in Figure 4. Before discussing the results, I note that it is reassuring that *trends* in Ohio and synthetic Ohio have similarities even in the post-implementation period. For example, looking at panel (a) on enrollment level of Black enrollment, note that enrollment level dropped in 2015 and increased the year after in both Ohio and synthetic Ohio. Trends in enrollment level of white students are also quite similar for Ohio and synthetic Ohio, although levels differ for most demographic groups. This is suggestive evidence that the synthetic controls proxy for a counterfactual Ohio well.

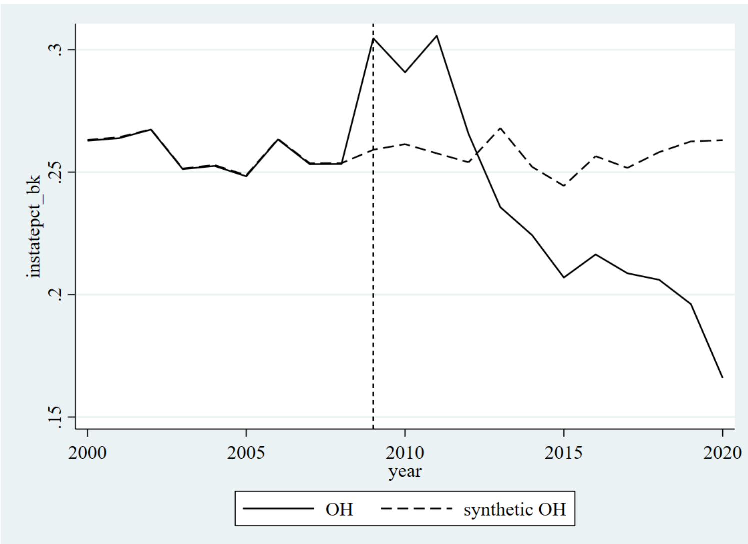
Overall and for all demographic groups, enrollment level in Ohio began increasing relative to enrollment level in synthetic Ohio starting in 2009. Enrollment levels remained higher than synthetic Ohio until 2013-2014; after that, in all demographic groups, enrollment level decreased relative to that in synthetic Ohio.[25] Summing up the per-period changes, I estimate that the long-run changes in enrollment level are: a 1 percentage point decrease overall (p-value = 0.068), a 2.8 percentage point decrease for Black students (p-value = 0.028), a 1.5 percentage point decrease for Hispanic students (p-value = 0.111), and a 0.4 percentage point decrease for white students (p-value = 0.157).

The timing of these changes coincides well with funding policy changes: overall and white enrollment levels increased around the timing of the initial change (which increased funding), but leveled off between 2013-2014 in anticipation of the fact that the FY 2014 changes decreased expected funding. This analysis is therefore consistent with Hypothesis 1 that universities did not respond to a change in the structure of the funding rule; rather, they responded to changes in how much funding they expected to receive.

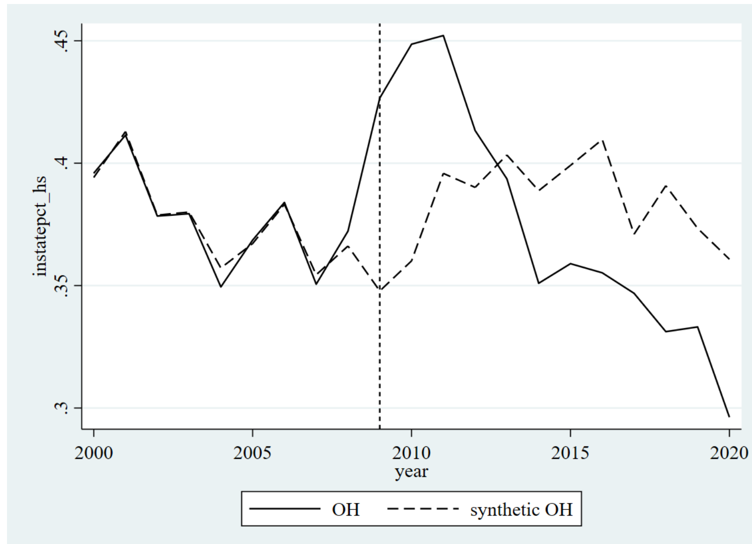
[25] While enrollment level of white students and overall enrollment level are relatively stable between 2013-2019, note that this comparison should be made to *synthetic Ohio*. Overall and white enrollment levels continued increasing for synthetic Ohio between 2009-2019, whereas overall and white enrollment level stops increasing after 2013 in Ohio.

Figure 4: Enrollment Levels, Overall and by Demographic Group

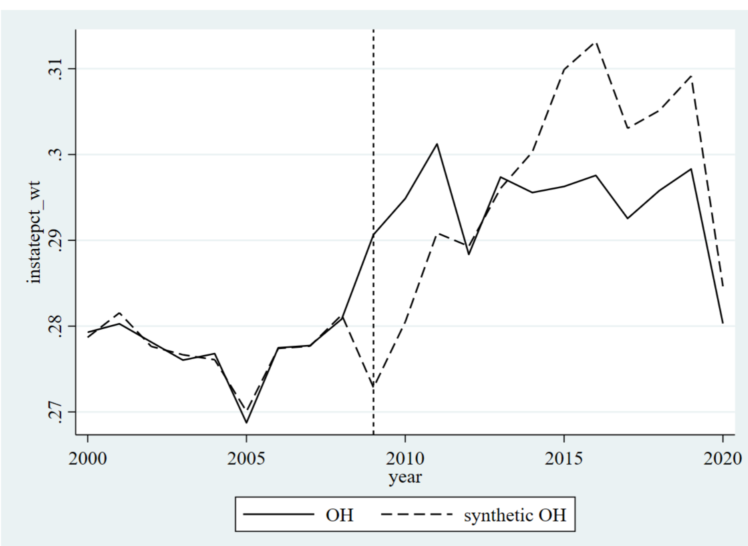
(a) Black



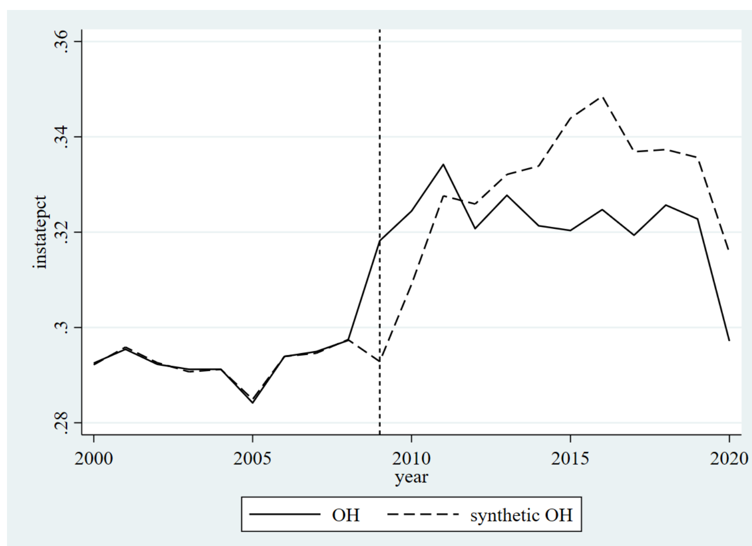
(b) Hispanic



(c) White, Adjusted



(d) Overall, Adjusted



The differences between demographic groups also merits discussion. White enrollment level for Black and Hispanic students have similar trends, enrollment level for white students appears to respond differently. Starting around 2013-2014, enrollment level decreases for all demographic groups relative to synthetic Ohio. However, for Black and Hispanic students, the decrease in enrollment level continues and even increases in magnitude over time. In contrast, enrollment level of white students levels off a few years after implementation. This is suggestive evidence that URM student enrollment was disproportionately affected by changes in funding, which will be further explored looking at changes in demographic composition.

6.2 Proportion of Black enrollment

In Figure 5, I show compositional changes in enrollment by demographic group. The four demographic groups I consider are Asian, Black, Hispanic, and white. However, the main outcome of interest is proportion of Black students, which I show in panel (b).

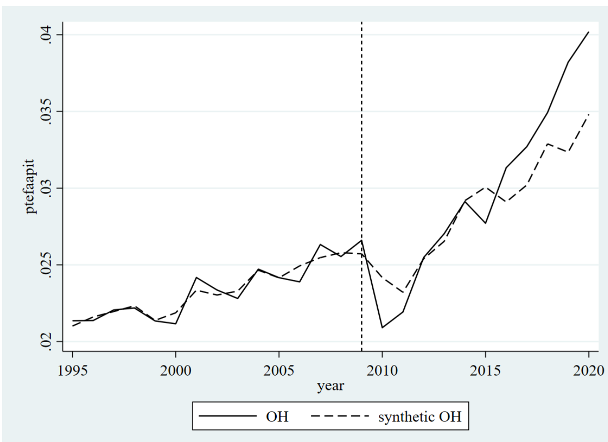
Proportion of Black enrollment increased for the first 3 years after implementation before sharply dropping around 2012. Afterwards, the proportion of Black students enrolled is persistently lower in Ohio compared to synthetic Ohio. While synthetic Ohio also experiences a long-term decrease in the proportion of Black enrollment, the magnitude of the drop is larger in Ohio. Interestingly, synthetic Ohio and Ohio appear to roughly move in parallel even in the post-period treatment except for the large drop in proportion of Black enrollment in Ohio that happened between 2012-2015, which could possibly suggest that Ohio universities learned from experience after being exposed to the new funding rules for a few years and made strategic changes in long-run enrollment policies. I estimate a long-run decrease in the proportion of Black enrollment of 1.13 percentage points (p-value = 0.059).

This analysis is consistent with Hypothesis 2: Black enrollment was disproportionately by changes in the funding rule. The proportion of Black enrollment initially and disproportionately increases between 2009-2011—this exactly the timeframe during which the public universities that enroll the most at-risk students stood to benefit the most from changes in funding rule as discussed in Section 3. There is then a sharp, disproportionate drop in proportion of Black enrollment between 2012-2015. During that entire time period, the importance of completions to funding was increasing. Also, in FY 2014, the state removed most of the policy elements that most benefited public universities who enroll at-risk students. Putting it altogether, proportion of Black enrollment appears to respond disproportionately both to expected increases and expected decreases in funding.

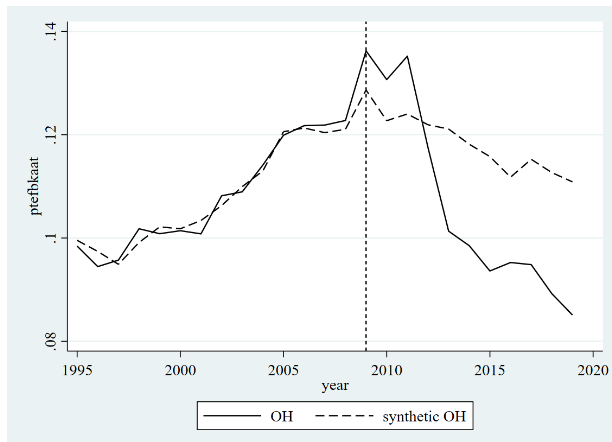
Finally, note that panel (a) of Figure 4 showing enrollment level of Black students and panel (b) of Figure 5 on proportion of Black students have different implications. While the results are similar, the pre-treatment

Figure 5: Proportion of Enrollment by Demographic Group

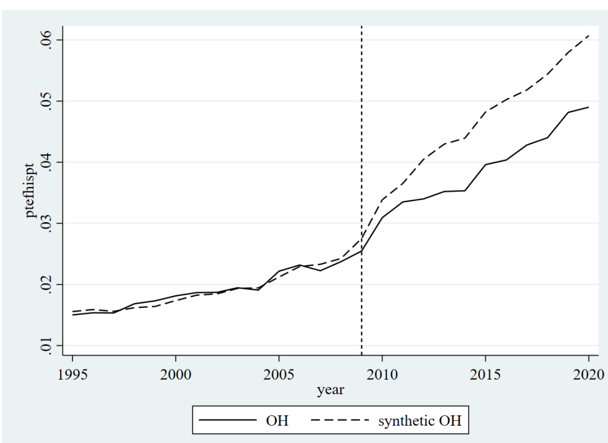
(a) Asian



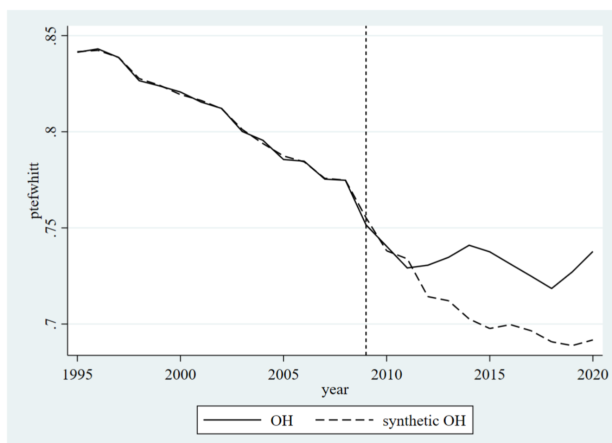
(b) Black



(c) Hispanic



(d) White



trends for the two outcomes are different. While enrollment level for Black students is fairly constant for both Ohio and synthetic Ohio during the pre-treatment period—and remains constant for synthetic Ohio post-treatment—the *proportion* of Black enrollment was increasing throughout the pre-treatment period. Since the number of Black 12th graders is relatively constant, this implies that the number of Black 12th graders who applied to public universities has been increasing over time, and public universities may have been expanding enrollment of Black students to respond to increased demand for higher education from Black students. However, after PF implementation, the number of potential Black applicants became much less predictive of how many Black students would go on to eventually enroll (although it remained predictive in synthetic Ohio, as seen by the fact that synthetic Ohio is fairly steady across the post-period treatment in panel (b) of Figure 4). This is further suggestive evidence that Black enrollment was indeed disproportionately affected by changes in funding.

I do not find statistically significant changes in proportion of enrollment for other demographic groups. Graphical inspection of panel (c) in Figure 5 suggests that Ohio universities admitted proportionately fewer Hispanic students following funding changes, but again, the change is not statistically significant. This may reflect differences between Hispanic and Black applicants as discussed in Section 5, or the fact that Ohio has a smaller population of Hispanic residents compared to the average state. The lack of statistical significance for other demographic groups implies that the changes in funding did not generate any positive preferential effects—that is, the sharp and disproportionate decrease in Black enrollment between 2012-2015 was not "absorbed" to the benefit of any specific demographic.

7 Policy Discussion and Conclusion

In this paper, I present a theoretical framework of state funding under general assumptions on how state policymakers and universities evaluate the returns to enrolling students to empirically evaluate the two main enrollment arguments for and against performance funding: that it (1) increases efficiency in enrollment, and (2) disproportionately affects enrollment of URM students.

The theoretical framework predicts that performance funding is not inherently more efficient in enrollment (Theorem 1), which goes against main argument (1) in favor of performance funding. Rather, the theory predicts that public universities respond to the amount of funding they expect to receive. I then show empirical evidence that this is the case, because Ohio enrollment levels increased during the time period in which funding was increased, and decreased during times when funding was expected to decrease. If state policymakers are concerned with inefficient over-enrollment, then the state should decrease overall funding. That said, switching to performance funding or increasing the emphasis on performance funding elements may be more politically viable than announcing a cut in funding, as it is possible to use the switch as an opportunity to decrease funding without directly calling it a funding cut.

My synthetic control analysis on proportion of Black enrollment provides evidence for argument (2) against performance funding. When looking at changes in demographic composition, I show in Figure 5 that enrollment of Black students appears to disproportionately increase in response to funding increases and disproportionately decrease in response to expected funding decreases. The theoretical framework predicts that this means Ohio universities likely under-enroll URM students. Specifically, Result 2 shows that public universities disproportionately change enrollment of URM students in response to funding changes if and only if the university's marginally accepted URM student gives the university *higher* private returns than the marginally accepted non-URM student. This suggests Ohio public universities are risk averse over enrolling of URM students, perhaps because URM students have lower completion rates on average. However, it is important to note that the theoretical framework also pushes against argument (2). In theory, performance funding is not *intrinsically* targeted against URM students. Issues in implementation, though, may disproportionately affect URM students.

This paper shows that there can be unintended and inequitable side effects to state policies that do not intend to affect equity. Ohio policymakers explicitly recognized that URM students are less likely to graduate by giving universities an additional incentive to admit URM students by including a subsidy for URM degree completions. In spite of this, the enrollment level of Black students was disproportionately affected by the change in funding rules; a back-of-the-envelope calculation suggests that this resulted in approximately 135 fewer Black students enrolled as first-time students at public four-year universities in Ohio from 2010-2020. A policymaker who cares for equity in college accessibility must be aware of *all* systematic differences between URM and non-URM students, not just differences in graduation rates, to correctly incentivize universities to change their enrollment decisions.

Two key takeaways are that universities respond to level of expected funding rather than the structure of a funding rule, and that level of funding may have unintended and inequitable effects on enrollment of students who are already historically under-represented. A secondary takeaway is that funding rules ought to be implemented at the institution level rather than at the state level. Finally, when designing enrollment incentives, state policymakers must holistically consider the costs and benefits that each university faces when deciding which students to admit and create incentives that holistically consider the gap between policymaker goals and institutional missions, rather than looking only at differences between groups of students. The modeling framework in this paper can be used to help guide the creation of straightforward, transparent funding rules and close gaps between theory and practice.

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A Theoretical Results

A.1 Cut-offs Under Binding Capacity Constraints

In the theoretical framework, I assume capacity constraints do not bind at public universities. This is not a realistic assumption, but socially efficient cut-offs on academic preparedness for each group can be derived with the following procedure:

1. Take some arbitrarily small $\varepsilon > 0$. Pick the group with the higher expected social returns at the academic preparedness level $1 - \varepsilon$, and admit all students in that group with academic preparedness at least as high as $1 - \varepsilon$.

2. Compare the expected social returns to admitting URM vs. M students at the top ε of academic preparedness who have not yet been admitted and admit students in that group.

3. Repeat until capacity constraints are met.

Using this procedure to derive socially efficient cut-offs does not change theoretical results.

A.2 The race-blind social planner

Call the SP "race blind" if $Educ_k^{SP}(a|M) = Educ_k^{SP}(a|URM)$ for all levels of academic preparedness a and analogously for $Degree_k^{SP}()$. If the SP is race blind, then the the SP is more selective over URM students.

Proposition 2. *Let the SP be race-blind. Then* $a_{URM}^{SP} \geq a_M^{SP}$.

Proof. Suppose capacity constraints do not bind. The two socially efficient cut-offs solve $EV_k^{SP}(a|g) = 0$, so

$$\begin{aligned} &(1 - Early(a_M^{SP}|M))[Educ_k^{SP}(a_M^{SP}) - c_2] + (1 - Late(a_M^{SP}|M))Degree_k^{SP}(a_M^{SP}|M) \\ = &(1 - Early(a_{URM}^{SP}|URM))[Educ_k^{SP}(a_{URM}^{SP}) - c_2] + (1 - Late(a_{URM}^{SP}|URM))Degree_k^{SP}(a_{URM}^{SP}). \end{aligned}$$

Now suppose for a contradiction that $a_{URM}^{SP} < a_M^{SP}$. It follows that $Early(a_{URM}^{SP}|URM) > Early(a_M|URM) > Early(a_M|M)$, and similarly for the late exit functions. For equality to hold it must be the case that at least one of the two conditions hold: (1) $Educ_k^{SP}(a_M^{SP}|M) = Educ_k^{SP}(a_M^{SP}|URM) < Educ_k^{SP}(a_{URM}^{SP}|URM)$ or (2) $Degree_k^{SP}(a_M^{SP}|M) = Degree_k^{SP}(a_M^{SP}|URM) < Degree_k^{SP}(a_{URM}^{SP}|URM)$. But neither condition can hold because $Educ_k^{SP}()$ and $Degree_k^{SP}()$ are increasing in academic preparedness and $a_{URM}^{SP} < a_M^{SP}$. Hence the contradiction.

Now suppose capacity constraints bind. Again suppose for a contradiction that $a_{URM}^{SP} < a_M^{SP}$. But then it must be the case that the social returns *all* the URM students with academic preparedness $a \in [a_{URM}^{SP}, a_M^{SP}]$

yield lower social returns to enrollment than the M student at the M-type academic preparedness, contradicting the admissions procedure with capacity constraints. □

A.3 The Efficient Ohio PF Rule

Here I give the full statement of two results from Section 4.4. First, there exists a unique socially efficient Ohio PF rule.

Proposition 3. *A socially efficient Ohio PF rule exists. It is unique.*

Proof. By construction. First, for convenience of notation, denote

$$\Delta Educ_k(a_g|g) = Educ_k^{SP}(a_g|g) - Educ_k(a_g|g)$$
$$\Delta Degree_k(a_g|g) = Degree_k^{SP}(a_g|g) - Degree_k(a_g|g)$$

First, set (ϕ, p) to induce the optimal M cut-offs,

$$(1-\phi)p = \Delta Educ_k(a_M^{SP}|M),$$
$$\phi p = \Delta Degree_k(a_M^{SP}|M).$$

Then separate ψ into two components, $(\psi_1, \psi_2) \in \mathbb{R}^2$ such that $\psi = \psi_1 + \psi_2$, and set them to satisfy

$$\phi\psi_1 p = \frac{1 - Early(a_{URM}^{SP}|URM)}{1 - Late(a_{URM}^{SP}|URM)}(\Delta Educ_k(a_{URM}^{SP}|URM) - (1-\phi)p)$$
$$\phi\psi_2 p = \Delta Degree_k(a_{URM}^{SP}|URM) - \phi p.$$

The ψ_1 component realigns k's problem at $t = 1$ with the SP's problem, and analogously for ψ_2. Solving for each component,

$$\psi_1 = \frac{1 - Early(a_{URM}^{SP}|URM)}{1 - Late(a_{URM}^{SP}|URM)}[\frac{\Delta Educ_k(a_{URM}^{SP}|URM)}{\Delta Degree_k(a_M^{SP}|M)} - \frac{1-\phi}{\phi}]$$
$$\psi_2 = \frac{\Delta Degree_k(a_{URM}^{SP}|URM)}{\Delta Degree_k(a_M^{SP}|M)} - 1,$$

and implementing $(p, \phi, \psi_1 + \psi_2)$ aligns k's problem with the SP's problem at the efficient cut-offs. □

Recall that the actual subsidy for URM degree completions in Ohio is

$$\psi^{OH} = Late(a_{URM}^{SP}|URM) - Late(a_M^{SP}|M).$$

Except in a knife-edge case, it is unlikely that the *actual* Ohio PF rule implemented—which takes into account only the difference in graduation rates between groups of students—aligns with the socially efficient Ohio PF rule. This is because an efficient funding rule has to take into account the fact that the social returns to enrollment are systematically different from the private returns.

Corollary 2. *Let the proposed Ohio PF rule be efficient over M students. k's optimal URM cut-off is efficient if and only if*

$$\begin{aligned} Late(a^{SP}_{URM}|URM) - Late(a^{SP}_{M}|M) =& \frac{1 - Early(a^{SP}_{URM}|URM)}{1 - Late(a^{SP}_{URM}|URM)} \left[\frac{\Delta Educ_k(a^{SP}_{URM}|URM)}{\Delta Degree_k(a^{SP}_{M}|M)} - \frac{1-\phi}{\phi} \right] \\ &+ \frac{\Delta Degree_k(a^{SP}_{URM}|URM)}{\Delta Degree_k(a^{SP}_{M}|M)} - 1 \end{aligned}$$

For the sake of discussing whether or not the knife-edge case is likely (or put another way, if the URM subsidy ψ is likely to serve its intended purpose), note that two of the components in Corollary 2 can be approximated.

First, consider the component $\frac{1-Early(a^{SP}_{URM}|URM)}{1-Late(a^{SP}_{URM}|URM)}$. While we cannot know these statistics at the socially efficient cut-offs, *Snapshot Report: First-Year Persistence and Retention* (2017) and Causey, Pevitz, Ryu, Scheetz, and Shapiro (2022) show that for the Fall 2015 cohort, the first year persistence rate for Black students at four-year public universities was 64.8%, while the six-year graduation rate was 51.3%. So an estimate for this component is approximately 1.26.

Second, consider $\frac{1-\phi}{\phi}$. In Ohio, the weight on course completion is 30% and degree completion is 50%. Disregarding set-aside funds, that implies $\phi = 0.625$ ($\frac{1-\phi}{\phi} = 0.6$).

Although the components

$$\frac{\Delta Educ_k(a^{SP}_{URM}|URM)}{\Delta Degree_k(a^{SP}_{M}|M)}, \frac{\Delta Educ_k(a^{SP}_{URM}|URM)}{\Delta Degree_k(a^{SP}_{M}|M)}$$

can't be approximated without imposing further assumptions, such as a precise functional form, understanding how they affect the size of the optimal URM subsidy is important for characterizing why ψ^{OH} may be incorrectly specified. Recalling that $EV^{SP}_k(a_i|g_i) > EV_k(a_i|g_i)$ for all types by assumption, the size of the right-hand side of Corollary 2 depends on how large $\Delta Educ_k(a^{SP}_{URM}|URM) > 0$ and $\Delta Degree_k(a^{SP}_{URM}|URM) > 0$ are relative to $\Delta Degree_k(a^{SP}_{M}|M) > 0$. Fixing $\Delta Degree_k(a^{SP}_{M}|M)$, as the difference in how the SP and the university k value URM enrollment becomes larger, the optimal subsidy becomes larger. On the other hand, fixing the numerators, as the difference in how the SP and university k value M enrollment becomes larger, the optimal subsidy becomes smaller.

A.4 Full Comparative Statics

First, define $\bar{\psi}$ as:

$$\bar{\psi} = \frac{Late(a^*_{URM}|URM) - Early(a^*_{URM}|URM)}{1 - Late(a^*_{URM}|URM)}. \tag{6}$$

By construction, $\bar{\psi} > 0$.

The full comparative statics on how changes in funding rules are expected to change enrollment decisions follow.

Proposition 4. *Let $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M) > 0$. k's optimal M cut-off is decreasing in the level of funding p and increasing in the proportion of funding determined by degree completions ϕ.*

Let $\frac{\partial}{\partial x^*_{URM}} EV_k(x^*_{URM}|URM) > 0$. k's optimal URM cut-off is decreasing in p and ψ. Moreover, let $\psi \in [0, \bar{\psi}]$ ($\psi > \bar{\psi}$), where $\bar{\psi}$ is defined in Equation 6. k's optimal URM cut-off increasing (decreasing) in ϕ.

Proof. First, by construction, $EV_k()$ is weakly increasing in academic preparedness for both groups. For the M cut-off,

$$\begin{aligned}
\frac{\partial}{\partial p} EV_k(x^*_M|M) &= (1-\phi)(1 - Early(a^*_M|M)) + \phi(1 - Late(a^*_M|M)) \\
&> 0, \\
\frac{\partial}{\partial \phi} EV_k(a^*_M|M) &= p[Early(a^*_M|M) - Late(a^*_M|M)] \\
&< 0 \\
\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M) &\geq 0
\end{aligned}$$

so the Implicit Function Theorem gives that $x'_M(p)$ has the opposite sign of $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M)$, and $x'_M(\phi)$ has the same sign as $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M)$. Letting $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M) > 0$ hold, the results for the M cut-off comparative statics hold.

For the URM cut-off,

$$\begin{aligned}
\frac{\partial}{\partial p} EV_k(a^*_{URM}|URM) &= (1-\phi)(1 - Early(a^*_{URM}|URM)) + \phi(1+\psi)(1 - Late(a^*_{URM}|URM)) \\
&> 0, \\
\frac{\partial}{\partial \phi} EV_k(a^*_{URM}|URM) &= p[(1+\psi)(1 - Late(a^*_r|r)) - (1 - Early(a^*_{URM}|URM))] \\
\frac{\partial}{\partial \psi} EV_k(a^*_{URM}|URM) &= (1 - Late(a^*_{URM}|URM))\phi p \\
&> 0,
\end{aligned}$$

$$\frac{\partial}{\partial a^*_{URM}} EV_k(a^*_{URM}|URM) \geq 0,$$

so the Implicit Function theorem gives that $x'_{URM}(p)$ and $x'_{URM}(\psi)$ have the opposite sign of $\frac{\partial EV_k(a^*_M|M)}{\partial a^*_M}$. Letting $\frac{\partial}{\partial a^*_{URM}} EV_k(a^*_{URM}|URM) > 0$, the results for the URM cut-off comparative statics in p and ψ hold.

The sign of $\frac{\partial}{\partial \phi} EV_k(a^*_{URM}|URM)$ depends on how large ψ is, and is negative so long as $\psi \leq \bar{\psi}$. So if the subsidy is sufficiently small, then $x'_{URM}(\phi)$ has the same sign as $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M)$. If the subsidy is too large, the opposite is true. □

First, note that the conditions $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M) > 0$ and $\frac{\partial}{\partial a^*_{URM}} EV_k(a^*_{URM}|URM) > 0$ are not strongly restrictive and are satisfied so long as $EV_k()$ is not constant in academic preparedness around the university's cut-off a^*_g for both groups $g \in \{M, URM\}$. A stronger assumption on $EV_k()$ that guarantees the conditions hold is to let $EV_k()$ be strictly increasing in academic preparedness.

However, if the strictly increasing marginal academic preparedness conditions do not hold, then it is unclear how an increase in funding level p affects enrollment levels. For example, if capacity constraints bind, then an increase in p may actually decrease enrollment because students both at and just above the group-optimal cut-off have an expected return of 0, so k is no worse or better off by slightly increasing cut-offs. If capacity constraints are not binding, though, then the increase in p may still increase enrollment, because students just below the optimal cut-off before the funding increase have the same, strictly positive expected return as students at the cut-off.

Next, the size of the URM subsidy ψ determines how an increase in the fraction of funding determined by degree completions ϕ affects level of enrollment. If ψ is large and ϕ increases under the URM cut-off condition, then level of enrollment for URM students may increase in spite of the increased emphasis on degree completion. This is because a large ψ may distort the university's incentives if the "reward" for graduating URM students is so high that universities are willing to increase their exposure to risk and increases enrollment of URM students to potentially yield a high "reward" later on.

The second comparative statics result concerns whether changes in a funding rule disproportionately affect URM students.

Proposition 5. *Let* $\frac{\partial}{\partial a^*_M} EV_k(a^*_M|M) > 0$ *and* $\frac{\partial}{\partial a^*_{URM}} EV_k(a^*_{URM}|URM) > 0$. *A change in* p *induces a larger change in URM enrollment than M enrollment if and only if*

$$\frac{\frac{\partial}{\partial x} EV_k(a^*_M|M)}{\frac{\partial}{\partial x} EV_k(a^*_{URM}|URM)} < \frac{(1-\phi)(1-Early(a^*_{URM}|URM)) + \phi(1+\psi)(1-Late(a^*_{URM}|URM))}{(1-\phi)(1-Early(a^*_M|M)) + \phi(1-Late(a^*_M|M))}. \tag{7}$$

Proof. By the Implicit Function theorem, the magnitude of change in cut-offs caused by an increase in p is larger for the URM cut-off if and only if

$$\begin{aligned}
|x'_M(p)| &< |x'_{URM}(p)| \\
\frac{\frac{\partial}{\partial p}EV_k(a^*_M|M)}{\frac{\partial}{\partial p}EV_k(a^*_M|M)} &< \frac{\frac{\partial}{\partial p}EV_k(a^*_{URM}|URM)}{\frac{\partial}{\partial p}EV_k(a^*_{URM}|URM)} \\
\frac{\frac{\partial}{\partial x}EV_k(a^*_M|M)}{\frac{\partial}{\partial x}EV_k(a^*_{URM}|URM)} &< \frac{\frac{\partial}{\partial p}EV_k(a^*_{URM}|URM)}{\frac{\partial}{\partial p}EV_k(a^*_M|M)} \\
\frac{\frac{\partial}{\partial x}EV_k(a^*_M|M)}{\frac{\partial}{\partial x}EV_k(a^*_{URM}|URM)} &< \frac{(1-\phi)(1-Early(a^*_{URM}|URM)) + \phi(1+\psi)(1-Late(a^*_{URM}|URM))}{(1-\phi)(1-Early(a^*_M|M)) + \phi(1-Late(a^*_M|M))}.
\end{aligned}$$

□

Equation 7 is more likely to hold if (i) the private returns to the university to enrolling the marginal URM student is large relative to the private returns of enrolling the marginal non-URM student, (ii) the probability of URM persistence and retention at the URM cut-off is large relative to the probability of M persistence and retention at the M cut-off, or (iii) the URM subsidy ψ is large. Condition (ii) never holds if assuming that $a^*_{URM} < a^*_M$. This follows from Arcidiacono, Kinsler, and Ransom (2023), who show that URM enrollment at Harvard and University of North Carolina-Chapel Hill would become more selective if admissions were based on test scores alone. It is unclear if condition (iii) holds in Ohio, though discussion in the main analysis suggests this is likely the case.

Continuing to assume that $a^*_{URM} < a^*_M$, so that the right-hand side of Equation 7 is strictly less than 1, whether the inequality holds first depends on if the expected private returns to k is convex or concave. If $EV_k()$ is convex, then the marginal return to the cut-off URM student is always smaller than that of the cut-off M student, so the left-hand side of Equation 7 is greater than 1 and the inequality never holds. If $EV_k()$ is instead concave, then the left-hand side of Equation 7 is strictly less than 1. Concavity alone is not enough, though; the inequality might or might not hold depending both on *how* concave $EV_k()$ is as well as the difference in cut-offs $a^*_M - a^*_{URM}$.

Proposition 5 provides guidance for future work on estimating the returns to education, especially for researchers interested in the equity effects of policy changes on enrollment. There are many obstacles to this estimation problem, especially if the goal is to completely characterize $EV_K()$, because universities are all selective to some degree. It may be difficult, for example, to estimate the shape of the returns to education for students at a given university whose academic preparedness is much lower than the average student at the university—because it is unlikely that such a student would ever be accepted at the university to begin with. However, an advantage of Proposition 5 is that it only requires that the researcher estimate the returns to education around the existing cut-offs at a given university. Therefore, the researcher may be able to take advantage of policy changes at both the state and institution level that cause small changes in level of enrollment—this alone is enough to make predictions on the equity effects of policy changes, allowing the researcher to side-step the much more onerous task of fully estimating $EV_k()$.

B Alternate Candidate Pools

When constructing the synthetic controls, I considered the following candidate pools for the main analyses:

Table 5: Alternate Candidate Pools

1	Drop only the other treatment state, AK, and HI
2	Additionally drop CA, TX
3	Additionally drop Far West and New England states
4	Additionally drop Southwest states[26]

The candidate pool that I ultimately used for the main analyses is group 3. Candidate pools 1 and 2 result in synthetic controls that assign more weights than there are pre-treatment outcomes to match on for most of the enrollment level outcomes, hence I only present candidate pool 4 for those outcomes.

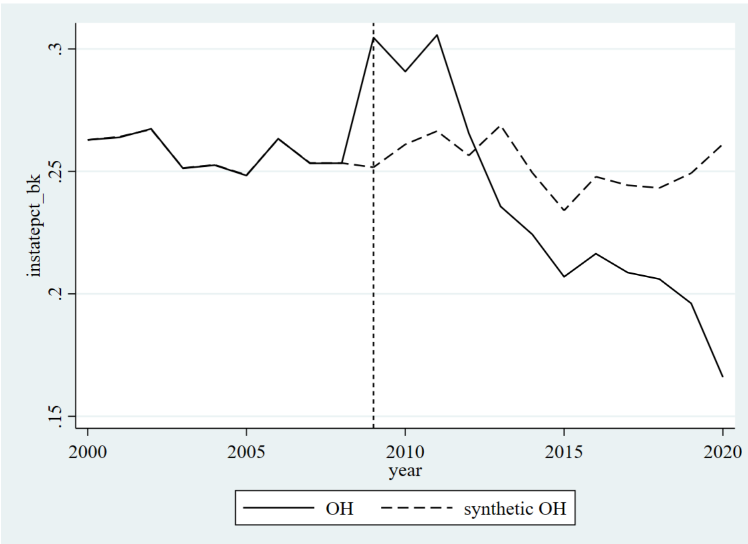
While the synthetic control weights change for virtually all enrollment level outcomes, I still find a statistically significant divergence in selectivity over Black enrollment (p-value = 0.03) in Ohio that follows the same trend as when I use candidate pool 3, though per-period treatment effects are muted when compared to the main estimation. Also, the per-period post-implementation treatment effects on enrollment for Hispanic students in Ohio is similar to the main estimation, but the p-value is lower (p=0.061). Including inference, using candidate pool 4 to estimate changes in overall enrollment level and enrollment level of white students (after treating the one-time accounting-related drop in the population of white 12th grade students) gives similar results to those presented in the main paper.

Looking at changes in enrollment composition, I find that the candidate pool affects only inference. All 4 candidate pools generate the same synthetic control when estimating changes in proportion of Black enrollment, so I have not included figures for these estimations. The p-values are 0.125 for candidate pool 1, 0.087 for candidate pool 3, and 0.059 for candidate pool 4.

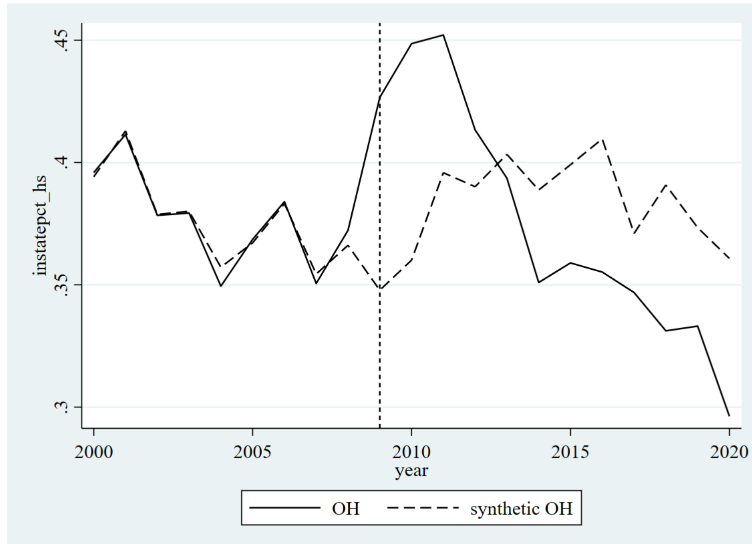
[26] Using the IPEDS classification, Southwest states include Arizona, New Mexico, Oklahoma, and Texas.

Figure 6: Synthetic Control: Enrollment Level, Ohio - Candidate Pool 4

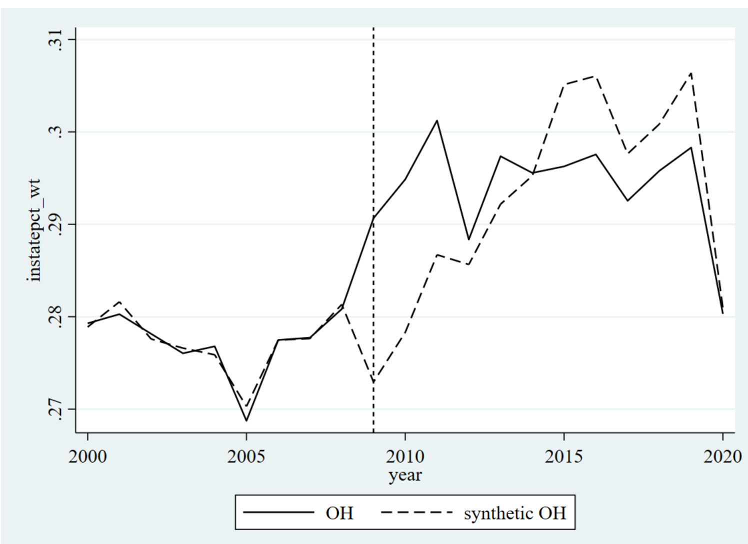
(a) Black



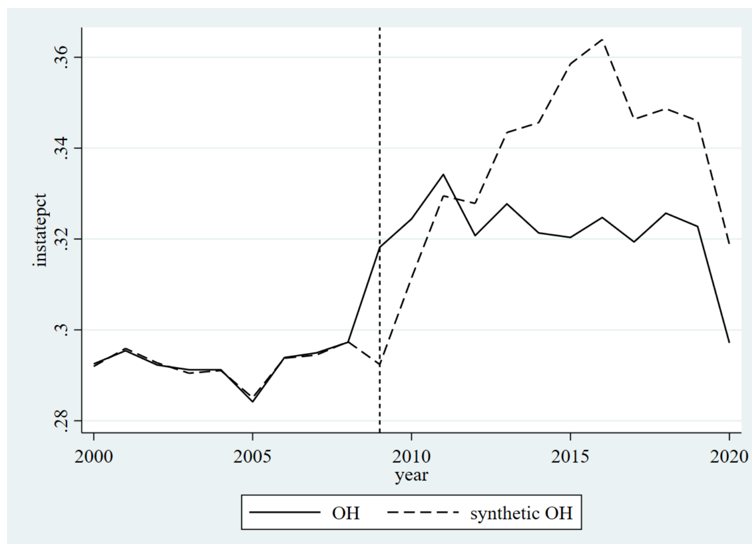
(b) Hispanic



(c) White - Modified



(d) Overall - Modified



C Alternate Matching Specifications

I follow recommendations from Ferman, Pinto, and Possebom (2020) and run alternate matching specifications to create the synthetic control, using my most preferred candidate pool throughout. I consider up to 4 alternate specifications: even and odd years only, both with and without additional controls (five year average of state unemployment rate and mean in-state tuition before implementation).[27] Data on state unemployment rates are from the U.S. Bureau of Labor Statistics; data on in-state tuition are from IPEDS.

I do not show from using even/odd years only because they result in synthetic controls that assign more weights than there are pre-treatment outcomes. In the interest of brevity, I also do not show overall enrollment level and enrollment level of white students in Ohio due to the accounting change that affected how 12th grade white students were counted, but the associated figures and tables may be released upon request. In Figure 7, I show the results on enrollment level using either even or odd years with additional controls for enrollment level.

In general, pre-treatment match quality is worse when discarding some of the pre-treatment outcomes as matching variables. This is not surprising, because these alternate matching specifications deliberately use fewer matching variables. While overall patterns in outcomes still hold, inference substantially degrades, likely because of the worse pre-treatment match quality. The p-values for Black enrollment level vary between 0.056 to 0.083 (evens and controls) but goes as high as 0.194 for odds and controls. The p-values for enrollment level of Hispanic enrollment vary between 0.028 to 0.083 (evens and controls) but goes as high as 0.222 for odds and controls.

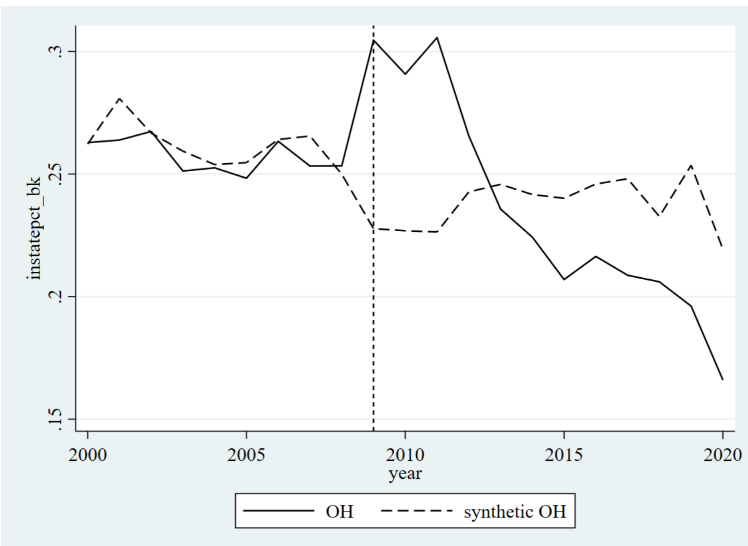
In Figure 8, I show the results on proportion of Black enrollment using either even or odd years with additional controls for enrollment level. The specification using only even years is fairly close to the specification using all pre-treatment years to match on; the p-value for both even years specifications is 0.054. Match quality for specifications using only odd years is worse, and the poorer match quality causes Ohio to perform worse in the standard placebo tests used for inference with synthetic controls (the lowest p-value for odd year specifications is 0.162).

However, for all outcomes checked, changing the matching variables does not qualitatively affect the trend in how the synthetic controls diverge from outcomes in the PF states post-implementation.

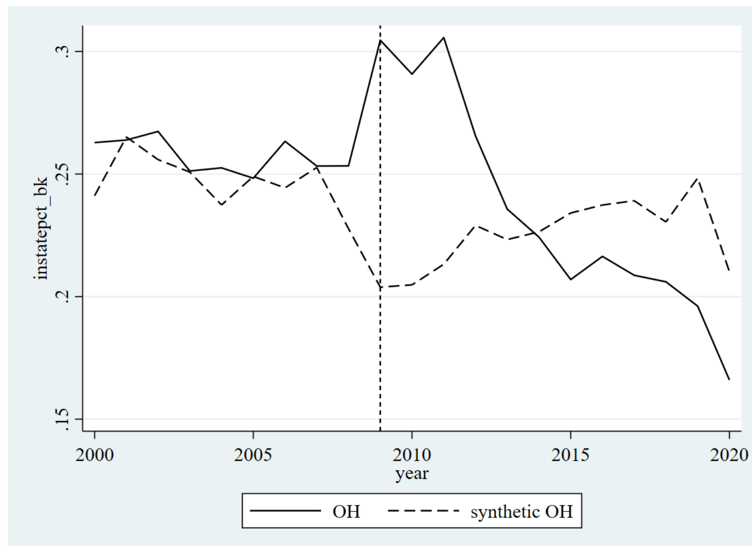
[27] Ferman, Pinto, and Possebom (2020) additionally suggest other specifications that are not well-suited for this analysis; for example, they recommend the first half of pre-treatment outcome variables—but this leads to synthetic controls that do not match with the treatment units very well in the second half of the pre-treatment period, and I believe that capturing the trend right before treatment occurs is important.

Figure 7: Synthetic Control: Enrollment Levels - Ohio, Alternate Matching Variables

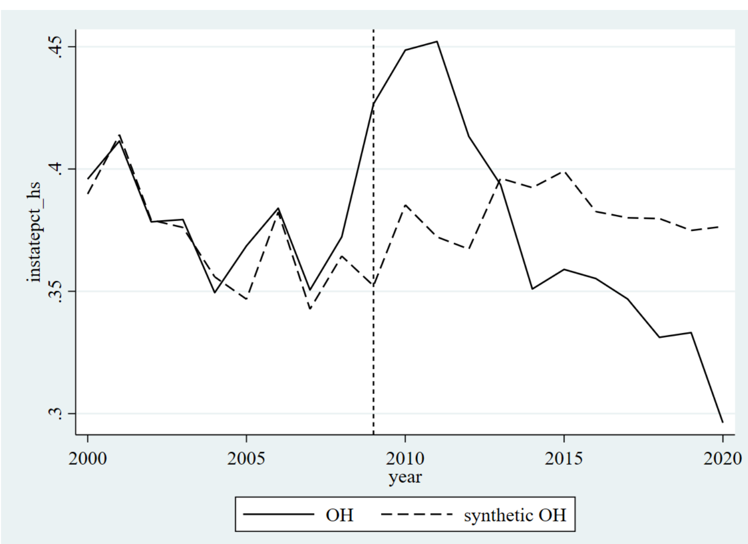
(a) Black Enrollment - Evens + Controls



(b) Black Enrollment - Odds + Controls



(c) Hispanic Enrollment - Evens + Controls



(d) Hispanic Enrollment - Odds + Controls

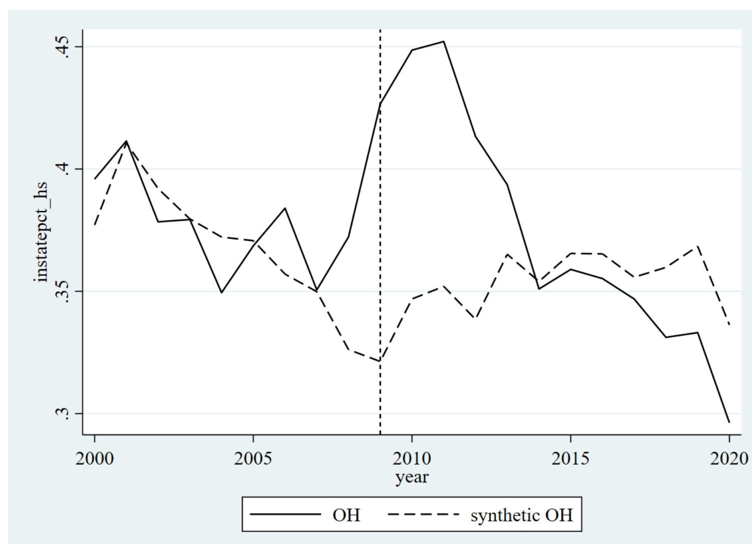
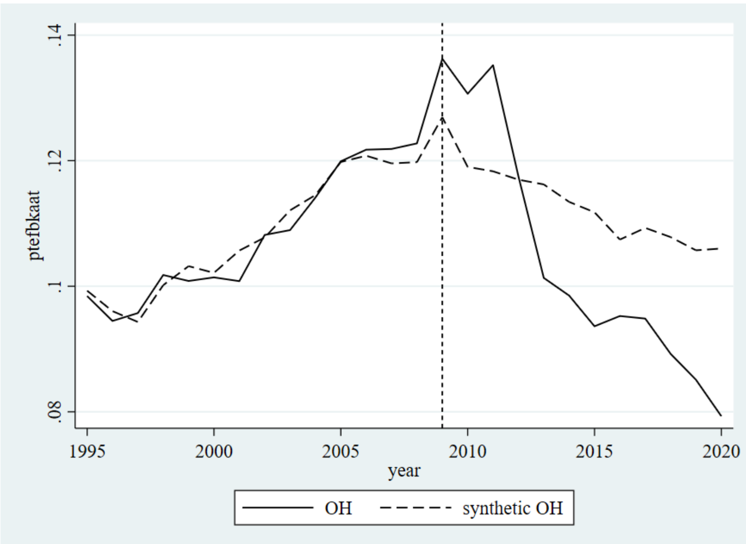
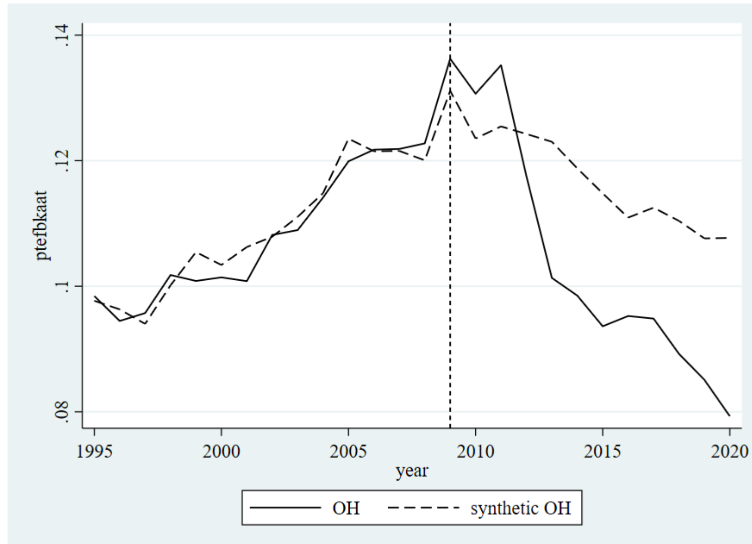


Figure 8: Synthetic Control: Proportion of Black Enrollment, Ohio - Alternate Matching Variables

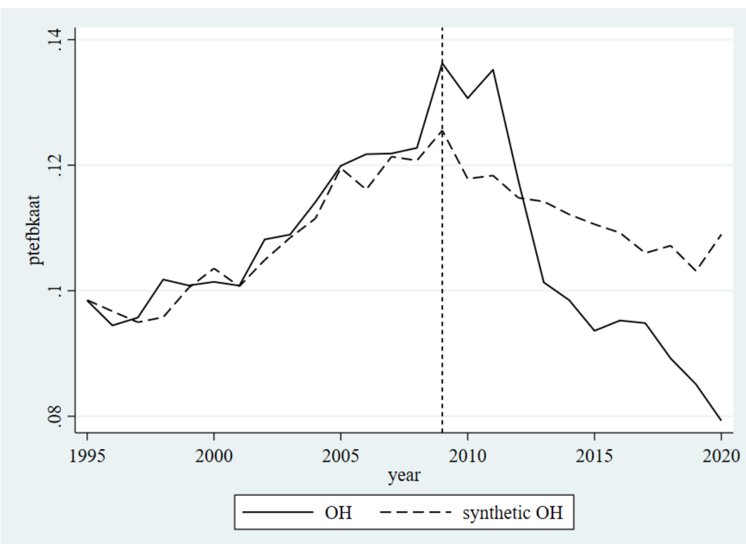
(a) Evens



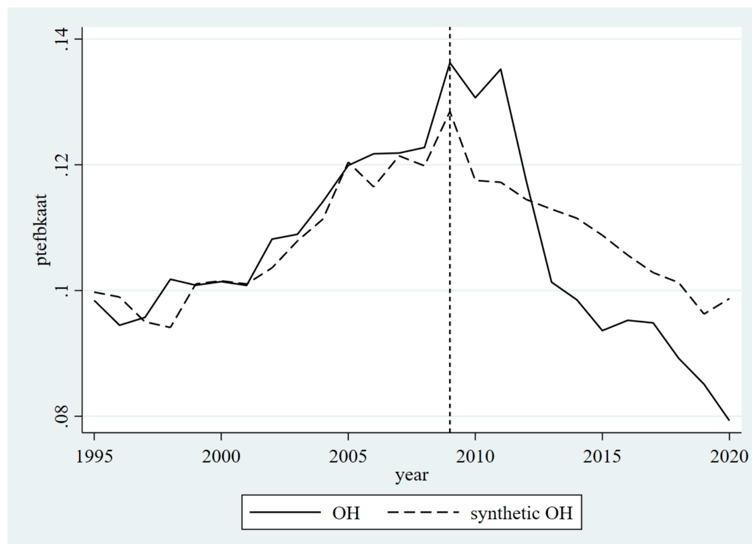
(b) Evens + Controls



(c) Odds



(d) Odds + Controls



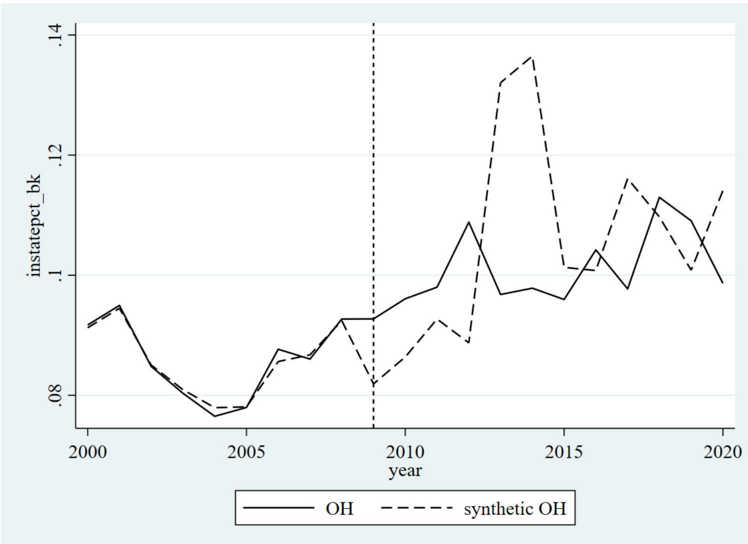
D Robustness Check: Private Universities

As a robustness check on the main empirical analysis as outline in Section 5, I performed the same analyses using enrollment at private four-year not-for-profit universities. These private universities are exposed to the same conditions as public universities in the same state *except* for public funding policies. If results in Section 6 are purely driven by statewide trends, then we should see similar results for public and private universities. For this robustness check, I use the same candidate pool as in the main analysis, but I additionally drop Wyoming due to not having candidate institutions in all years. I match on all pre-implementation incomes, as in the main estimations.

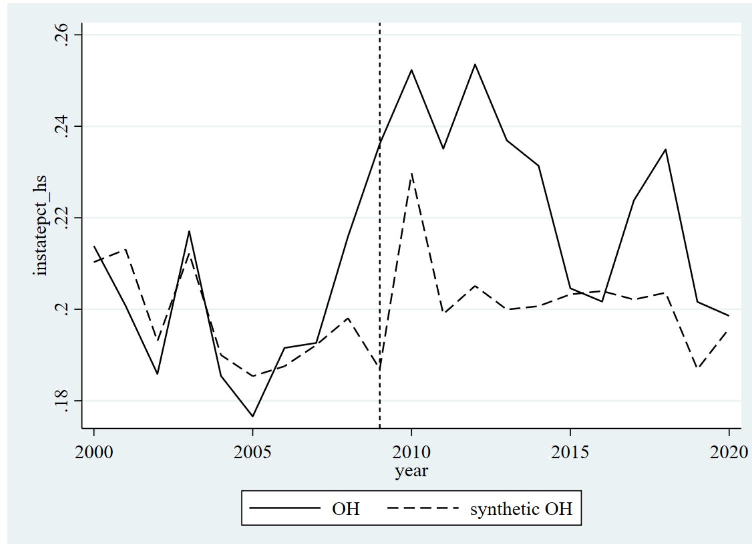
I find no evidence of similar changes in enrollment after PF implementation, which supports that the main results correctly capture the effects of PF. Figure 9 shows overall level of enrollment and enrollment level by demographic group. Overall, the private universities (1) generally have different enrollment trends than public universities, and (2) there is no statistically significant evidence that private universities changed their enrollment behavior in 2009 or afterwards. One possible concern is that for level of Black student enrollment, synthetic Ohio exhibits a sudden spike in 2013-2014, but this likely reflects volatility in the synthetic control rather than a true divergence between Ohio and its synthetic control. Figure 10 additionally shows that the main composition result from the paper does not hold: there is no evidence that private universities in Ohio changed proportion of Black enrollment in 2009 or afterwards.

Figure 9: Synthetic Control: Enrollment Level - Private Universities

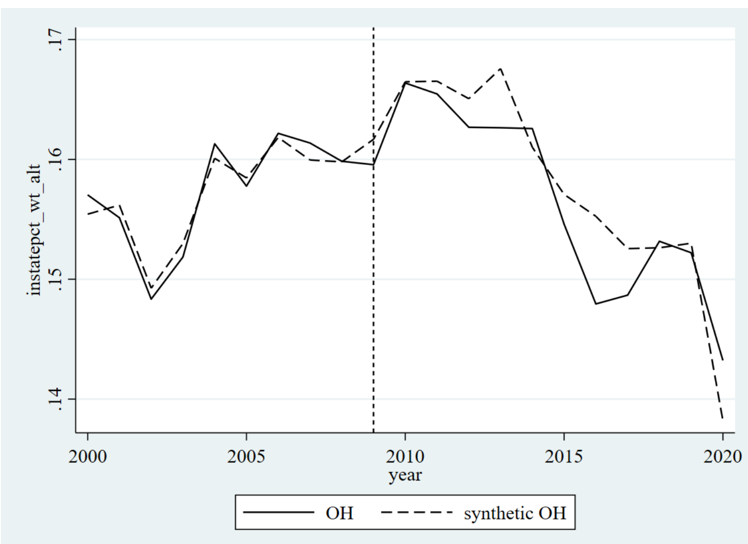
(a) Black



(b) Hispanic



(c) White (Modified)



(d) Overall (Modified)

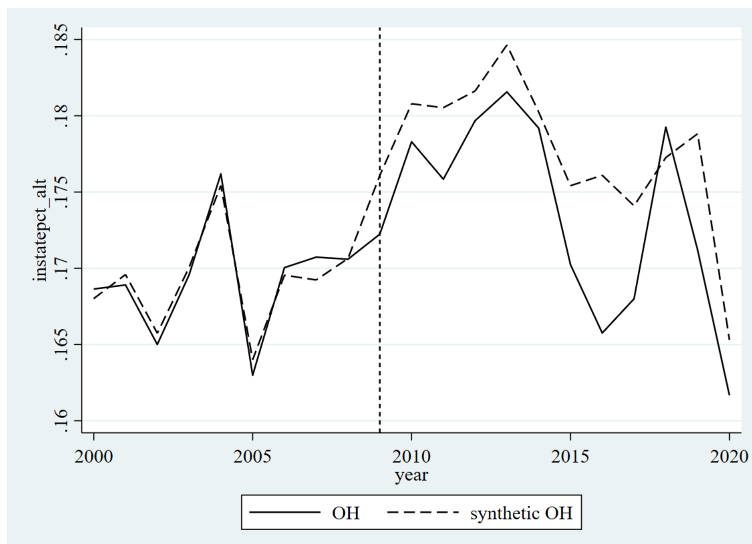
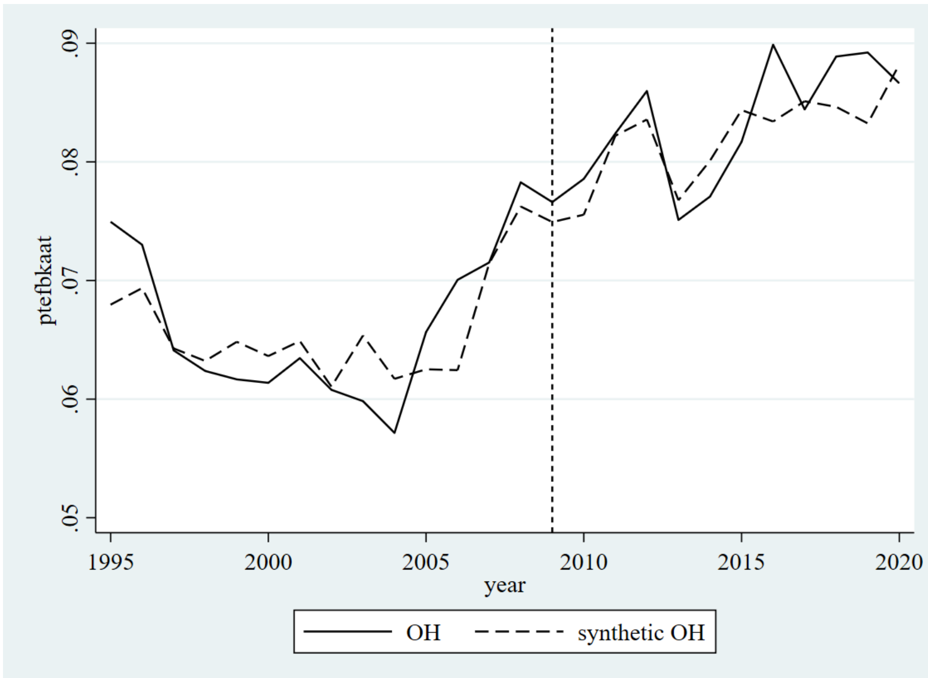


Figure 10: Synthetic Control: Proportion of Black Enrollment - Private Universities

Figure 11: Ohio



E Performance Funding in Tennessee

In the main analysis, I excluded Tennessee from analysis. This is because Tennessee has primarily used PF to appropriate funds to public universities since 2010. In principle, it is also possible to use Tennessee as a case study on implementation of PF, but analysis is complicated due to concurrent public policy changes.

The non-profit organization KnoxAchieves was started in 2008 with backing from local politicians to provide scholarship funds for low-income college freshmen in Knox County, and later expanded to the state level. The scholarship became state funded in 2014 when Tennessee legislators created the "Tennessee Promise" program. In its current iteration, the Tennessee Promise guarantees funding for all in-state high school graduates who enroll at a two-year program. Nguyen (2020) shows this caused short-run substitution between two-year and four-year institutions and indirectly increased enrollment at four-year institutions.

Information on Tennessee's public universities is summarized in Table 6. Tennessee has 10 public four-year universities, four of which are affiliated with each other through the University of Tennessee system. I exclude University of Tennessee-Southern, as it was a private institution until 2021. Tennessee public universities have less variance in size, proportion of in-state enrollment, and graduation rates compared to Ohio public universities. Overall, Tennessee universities are smaller and enroll proportionately more in-state

Table 6: Information on Tennessee Public Universities

Tennessee Universities	Enrollment	% In-State	6-Year Graduation %	Other Info
Austin Peay State University	8,120	88%	43%	Open admissions
East Tennessee State University	10,554	75%	55%	
Middle Tennessee State University	17,438	89%	55%	
Tennessee State University	7,678	38%	32%	Historically Black university, open admissions
Tennessee Technological University	8,537	94%	60%	
University of Memphis	16,708	81%	48%	
University of Tennessee-Chattanooga	9,884	89%	52%	Open admissions
University of Tennessee-Knoxville	27,039	54%	73%	Public flagship, has hospital
University of Tennessee-Martin	6,165	90%	53%	Open admissions

Data are from the NCES College Navigator. Enrollment is the total undergraduate population as of Fall 2022. 6-year graduation rate is for the Fall 2016 cohort.

Table 7: Summary Statistics: Mean Enrollment, Tennessee

	Non-PF States	Tennessee
Total Enrollment	16,781.47	17,884.38
	(15,260.24)	(2,410.32)
AAPI Enrollment	1,255.72	376.65
	(3,048.22)	(121.36)
Black Enrollment	1,868.77	3,495.08
	(2,131.39)	(515.24)
Hispanic Enrollment	1,873.47	494.50
	(4,597.56)	(364.04)
White Enrollment	10,454.28	12,642.69
	(7,133.27)	(1,082.29)

Number in parentheses are standard deviations. Tennessee is separated due to being another PF state, with additional discussion/analysis in Appendix E.

students than Ohio universities. Two exceptions are Tennessee State University and University of Tennessee-Knoxville, which likely enroll proportionately more out-of-state students due to being a historically Black university and the state flagship respectively.

Tennessee switched to PF in 2010 with a three year transition period similar that of Ohio. According to the *2015-2020 Outcomes-Based Funding Formula Overview* (2016), PF metrics differ across universities depending on institutional mission and Carnegie classification. Information on PF weights appears in Table 8. State policymakers assign weights with guidance from universities, which suggests that the incentive to respond to changes in funding rules is relatively weaker in Tennessee, as institutions should advocate for weights that align with pre-existing institutional characteristics. As suggestive evidence of this, note that actual 6-year graduation rates do not strongly correlate with the size of the PF weight on 6-year graduation rate. For example, East Tennessee State University and Middle Tennessee State University have similar characteristics, including 6-year graduation rates, but assign weights of 16.4% and 2.4% to 6-year graduation rates respectively. Almost all institutions place small weights on course completions and large weights on post-graduate degree completions, with Tennessee State University and the public flagship University of Tennessee-Knoxville additionally placing strong weights on research. There are only two at-risk characteristics considered across all metrics: age and financial status. In-state and out-of-state students are treated the same.

Table 8: Tennessee Performance Funding Weights

	24 Hours	48 Hours	72 Hours	Bachelors	Masters	Doctorates	Degrees/100 FTE	6-Year Grad %	Research
APSU	1.3%	2.5%	5.8%	44.3%	10.6%	23.4%	11.2%	0.0%	0.9%
ETSU	2.3%	3.7%	7.4%	29.0%	3.4%	20.0%	12.7%	16.4%	5.1%
MTSU	1.4%	2.9%	8.2%	42.9%	3.5%	21.7%	15.4%	2.4%	1.7%
TSU	1.5%	2.6%	6.2%	24.8%	6.7%	20.2%	10.5%	6.6%	21.1%
TTU	1.3%	2.9%	8.0%	32.9%	5.1%	37.9%	7.5%	0.7%	3.7%
UM	1.0%	2.1%	5.6%	29.9%	3.0%	22.5%	8.2%	20.4%	7.2%
UT-C	1.5%	2.9%	7.7%	31.2%	6.9%	39.0%	5.3%	2.9%	2.8%
UT-K	0.6%	1.7%	3.9%	19.2%	3.0%	18.2%	7.6%	21.9%	23.9%
UT-M	1.1%	2.1%	5.5%	27.5%	5.4%	55.2%	2.5%	0.0%	0.8%

As seen in Figure 12, the inflation-adjusted value of state appropriations in Tennessee have been comparable to average state appropriations in non-PF states, with some volatility from 2008-2012, after which state appropriations have been systematically lower than in the pre-PF period (i.e., 2010 and before). This suggests that level of state appropriations have been somewhat unpredictable in Tennessee compared to Ohio. The theory therefore predicts that there is a long-run decrease in enrollment level, especially from 2014 onward, but that this might be muted due to instability in expected funding and the fact that institutions choose their own funding weights.

Indeed, when I perform the same synthetic control analyses as in Section 5 using Tennessee as the treatment group, I do not find any results that are statistically significant at a p-value of 0.194 or lower.[28] As such, I do not show figures, but they are available on request. This empirical result is consistent with what the theory predicts, but weaker evidence than the case study of Ohio due to the confounding factors at play.

[28]Detailed analysis can be provided upon request.

Figure 12: Trends in State Appropriations, with Tennessee